order of bus impedance matrix is one, as shown below.

$$Z_{bus} = [j1.0]$$

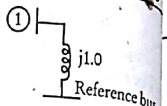



*Fig 1.18.2*

**Step 2 :** Connect bus-2 to bus-1 through an impedance j0.25 as shown in fig 1.18.3. This is case-2 modification and so the order of bus impedance matrix increases by one.

In the new bus impedance matrix, the elements of 1st column are copied as elements of 2nd column and the elements of 1st row are copied as elements of 2nd row. The diagonal element is given by $Z_{11} + Z_b$ where $Z_b$ = j0.25.

$$\therefore Z_{bus} = \begin{bmatrix} j1.0 & j1.0 \\ j1.0 & j1.0 + j0.25 \end{bmatrix} = \begin{bmatrix} j1.0 & j1.0 \\ j1.0 & j1.25 \end{bmatrix}$$

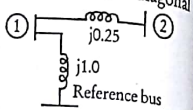



*Fig 1.18.3*

**Step 3 :** Connect the impedance j1.25 from bus-2 to reference bus as shown in fig 1.18.4. This is case-3 modification. In case-3 modification the new bus impedance matrix is formed as that of case-2 and then the last row and column are eliminated by node elimination technique.

In the new bus impedance matrix the elements of 2nd column are copied as elements of 3rd column and the elements of 2nd row are copied as elements of 3rd row. The diagonal element is given by $Z_{22} + Z_b$ where $Z_b$ = j1.25.

$$\therefore Z_{bus} = \begin{bmatrix} j1.0 & j1.0 & j1.0 \\ j1.0 & j1.25 & j1.25 \\ j1.0 & j1.25 & j1.25 + j1.25 \end{bmatrix}$$

$$= \begin{bmatrix} j1.0 & j1.0 & j1.0 \\ j1.0 & j1.25 & j1.25 \\ j1.0 & j1.25 & j2.5 \end{bmatrix}$$

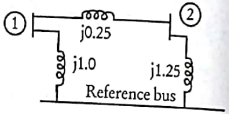



*Fig 1.18.4*

The actual new bus impedance matrix is obtained by eliminating the 3rd row and 3rd column. The element $Z_{jk}$ of the actual new bus impedance matrix is given by,

$$\therefore Z_{jk,act} = Z_{jk} - \frac{Z_{j(n+1)}\, Z_{(n+1)k}}{Z_{(n+1)(n+1)}} \qquad \text{where } n = 2 \text{ ; } j = 1, 2, \text{ and } k = 1, 2,$$

$$Z_{11,act} = Z_{11} - \frac{Z_{13}\, Z_{31}}{Z_{33}} = j1.0 - \frac{j1.0 \times j1.0}{j2.5} = j0.6$$

$$Z_{12,act} = Z_{12} - \frac{Z_{13}\, Z_{32}}{Z_{33}} = j1.0 - \frac{j1.0 \times j1.25}{j2.5} = j0.5$$

$$Z_{21,act} = Z_{12,act} = j0.5$$

$$Z_{22,act} = Z_{22} - \frac{Z_{23}\, Z_{32}}{Z_{33}} = j1.25 - \frac{j1.25 \times j1.25}{j2.5} = j0.625$$



$$\therefore Z_{bus} = \begin{bmatrix} j0.6 & j0.5 \\ j0.5 & j0.625 \end{bmatrix}$$

**Step 4 :** Connect the bus-3 to bus-2 through an impedance j0.05 as shown in fig 1.18.5. This is case-2 modification and so the order of the bus impedance matrix increases by one.

In the new bus impedance matrix, the elements of 2nd column are copied as elements of 3rd column and the elements of 2nd row are copied as elements of 3rd row. The diagonal element is given by $Z_{22} + Z_b$ where $Z_b$ = j0.5.

$$Z_{bus} = \begin{bmatrix} j0.6 & j0.5 & j0.5 \\ j0.5 & j0.625 & j0.625 \\ j0.5 & j0.625 & j0.625 + j0.05 \end{bmatrix}$$

$$\therefore Z_{bus} = \begin{bmatrix} j0.6 & j0.5 & j0.5 \\ j0.5 & j0.625 & j0.625 \\ j0.5 & j0.625 & j0.675 \end{bmatrix}$$

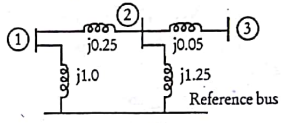



*Fig 1.18.5*

## 1.7. SYMMETRICAL COMPONENTS

The analysis of unsymmetrical polyphase network by the method of symmetrical components was introduced by *Dr. C. Fortesque*. He proved that an unbalanced system of *n* related vectors can be resolved into *n* system of balanced vectors called symmetrical components of original vectors. The *n* vectors of each set of components are equal in length and the phase angles between adjacent vectors of the set are equal.

In a three phase system, the three unbalanced vectors [either $V_a$, $V_b$ & $V_c$ or $I_a$, $I_b$ & $I_c$] can be resolved into three balanced system of vectors. The vectors of the balanced system are called symmetrical components of the original system. The symmetrical components of three phase system are,

1. Positive sequence components
2. Negative sequence components
3. Zero sequence components.

The positive sequence components consists of three vectors equal in magnitude, displaced form each other by 120° in phase, and having the same phase sequence as the original vectors

The negative sequence components consists of three vectors equal in magnitude, displaced from each other by 120° in phase, and having the phase sequence opposite to that of the original vectors.

The zero sequence components consists of three vectors equal in magnitude and with zero phase displacement from each other.

Let $V_a$, $V_b$ and $V_c$ be the set of unbalanced voltage vectors with phase sequence abc. Each voltage vector can be resolved into positive, negative and zero sequence components.

Let $V_{a1}, V_{b1}$ & $V_{c1}$ = Positive sequence components of $V_a$, $V_b$ & $V_c$ respectively with phase sequence abc.

$V_{a2}, V_{b2}, V_{c2}$ = Negative sequence components of $V_a$, $V_b$ and $V_c$ respectively with phase sequence acb.

$V_{a0}, V_{b0}$ & $V_{c0}$ = Zero sequence components of $V_a$, $V_b$ & $V_c$ respectively.

The vector diagrams of positive, negative and zero sequence components are shown in fig 1.13.

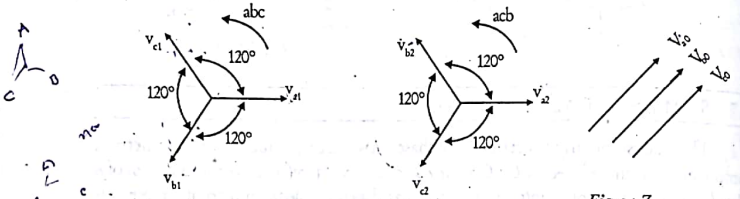



*Fig a : Positive sequence components*

*Fig b : Negative sequence components*

*Fig c : Zero sequence components*

*Fig 1.13 : Vector diagram of symmetrical components of unbalanced 3-phase voltage vectors.*

From the vector diagram of symmetrical components the following conclusions can be made.

1. On rotating the vector $V_{a1}$ by 120° in anticlockwise direction we get $V_c$
2. On rotating the vector $V_{a1}$ by 240° in anticlockwise direction we get $V_{b1}$.
3. On rotating the vector $V_{a2}$ by 120° in anticlockwise direction we get $V_{b2}$.
4. On rotating the vector $V_{a2}$ by 240° in anticlockwise direction we get $V_{c2}$.

Therefore, on rotating the symmetrical component of one vector by 120° or multiples of 120° we get the symmetrical components of other vectors. Hence we can define an operator which causes a rotation of 120° in the anticlockwise direction, such an operator is denoted by the letter "a".



The operator "a" is defined as,

$$a = 1\angle 120° = 1e^{+j2\pi/3} = \cos 2\pi/3 + j\sin 2\pi/3 = -0.5 + j0.866$$

Since, $a = 1\angle 120° = -0.5 + j0.866$ .....(1.83)

$a^2 = 1\angle 240° = -0.5 - j0.866$ .....(1.84)

$a^3 = 1\angle 360° = 1$ .....(1.85)

$1 + a + a^2 = 1 + (-0.5 + j0.866) + (-0.5 - j0.866) = 0$

$\therefore 1 + a + a^2 = 0$ .....(1.86)

## Computation of unbalanced vectors from their symmetrical components

Each of the original unbalanced vector is the sum of its positive, negative and zero sequence component. Therefore the original unbalanced three phase voltage vectors can be expressed in terms of their components as shown below.

$$V_a = V_{a0} + V_{a1} + V_{a2} \quad .....(1.87)$$

$$V_b = V_{b0} + V_{b1} + V_{b2} \quad .....(1.88)$$

$$V_c = V_{c0} + V_{c1} + V_{c2} \quad .....(1.89)$$

From the vector diagram shown in figure(1.11), we get the following relations between various symmetrical components.

$$V_{b0} = V_{a0} \; ; \quad V_{b1} = a^2 V_{a1} \; ; \quad V_{b2} = a V_{a2} \quad .....(1.90)$$

$$V_{c0} = V_{a0} \; ; \quad V_{c1} = a V_{a1} \; ; \quad V_{c2} = a^2 V_{a2} \quad .....(1.91)$$

Using equations (1.90) and (1.91), the equations (1.87) to (1.89) can be written as shown below.

$$V_a = V_{a0} + V_{a1} + V_{a2} \quad .....(1.92)$$

$$V_b = V_{a0} + a^2 V_{a1} + a V_{a2} \quad .....(1.93)$$

$$V_c = V_{a0} + a V_{a1} + a^2 V_{a2} \quad .....(1.94)$$

The equations (1.92) to (1.94) can be arranged in the matrix form as shown in equ(1.95).

$$\begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix} \quad .....(1.95)$$

The equation (1.95) can be used to compute the unbalanced voltage vectors from the knowledge of symmetrical components.

## Computation of symmetrical components of unbalanced vectors

The matrix equation (1.95) can be written in the vector notation as shown in equ(1.96)

$$V = A V_{sy} \qquad \text{.....(1.96)}$$

where, $V = \begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix}$ ; $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix}$ and $V_{sy} = \begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix}$

On premultiplying the equ(1.96) by $A^{-1}$ we get

$$A^{-1} V = V_{sy}$$

$$\therefore V_{sy} = A^{-1} V$$

$$A^{-1} = \frac{\text{Adjoint of A}}{\text{Determinant of A}} \qquad \text{.....(1.97)}$$

Let $\Delta$ = Determinant of A

$$\therefore \Delta = \begin{vmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{vmatrix} = 1 \times (a^4 - a^2) - 1 \times (a^2 - a) + 1(a - a^2) \qquad \left( \because a^4 = a^3 . a = a \right)$$

$$= (a - a^2) + (a - a^2) + (a - a^2) = 3(a - a^2)$$

Let $\Delta_{ij}$ = Cofactor of $A_{ij}$

$$\therefore \Delta_{11} = a^4 - a^2 = a - a^2$$

$$\Delta_{12} = -(a^2 - a) = a - a^2$$

$$\Delta_{13} = a - a^2$$

$$\Delta_{21} = -(a^2 - a) = a - a^2$$

$$\Delta_{22} = a^2 - 1$$

$$\Delta_{23} = -(a - 1) = 1 - a$$

$$\Delta_{31} = a - a^2$$

$$\Delta_{32} = -(a - 1) = 1 - a$$

$$\Delta_{33} = a^2 - 1$$

$$\therefore A^{-1} = \frac{1}{\Delta} \text{ Adjoint of } A = \frac{1}{\Delta} \begin{bmatrix} \Delta_{11} & \Delta_{12} & \Delta_{13} \\ \Delta_{21} & \Delta_{22} & \Delta_{23} \\ \Delta_{31} & \Delta_{32} & \Delta_{33} \end{bmatrix}^T$$

$$= \frac{1}{\Delta} \begin{bmatrix} \Delta_{11} & \Delta_{21} & \Delta_{31} \\ \Delta_{12} & \Delta_{22} & \Delta_{32} \\ \Delta_{13} & \Delta_{23} & \Delta_{33} \end{bmatrix}$$



$$A^{-1} = \frac{1}{3(a - a^2)} \begin{bmatrix} a - a^2 & a - a^2 & a - a^2 \\ a - a^2 & a^2 - 1 & 1 - a \\ a - a^2 & 1 - a & a^2 - 1 \end{bmatrix}$$

$$A^{-1} = \frac{(a - a^2)}{3(a - a^2)} \begin{bmatrix} 1 & 1 & 1 \\ 1 & \frac{a^2 - 1}{a - a^2} & \frac{1 - a}{a - a^2} \\ 1 & \frac{1 - a}{a - a^2} & \frac{a^2 - 1}{a - a^2} \end{bmatrix} \qquad \text{.....(1.98)}$$

$$\frac{a^2 - 1}{a - a^2} = \frac{a^2 - a^3}{a - a^2} = \frac{a(a - a^2)}{(a - a^2)} = a \qquad \text{.....(1.99)}$$

$$\frac{1 - a}{a - a^2} = \frac{a^2(1 - a)}{a^2(a - a^2)} = \frac{a^2(1 - a)}{(a^3 - a^4)} = \frac{a^2(1 - a)}{1 - a} = a^2 \qquad \text{.....(1.100)}$$

Note : $a^3 = 1$ ; $a^4 = a^3 . a = a$

Using equations (1.99) and (1.100) the equation (1.98) can be written as

$$A^{-1} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \qquad \text{.....(1.101)}$$

On substituting for $A^{-1}$ from equ(1.101) in equ(1.97) we get

$$V_{sy} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} V$$

$$\begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} \qquad \text{.....(1.102)}$$

The matrix equ(1.102) can be expressed as three independent linear equations as shown below.

$$\therefore V_{a0} = \frac{1}{3} \left[ V_a + V_b + V_c \right] \qquad \text{.....(1.103)}$$

$$V_{a1} = \frac{1}{3}\left[V_a + aV_b + a^2V_c\right] \quad \text{.....(1.104)}$$

$$V_{a2} = \frac{1}{3}\left[V_a + a^2V_b + aV_c\right] \quad \text{.....(1.105)}$$

The equations (1.102) to (1.105) can be used to compute the symmetrical components of the unbalanced voltages.

## Symmetrical components of unbalanced current vectors

The symmetrical components of unbalanced current vectors can be obtained by an analysis similar to that of voltage vectors. All the equations developed for voltages can be used for current if we replace V by I.

Let $I_a, I_b, I_c$ = Unbalanced current vectors with phase sequence abc.

$I_{a1}, I_{b1}, I_{c1}$ = Positive sequence components of $I_a, I_b$ and $I_c$ respectively with phase sequence abc.

$I_{a2}, I_{b2}, I_{c2}$ = Negative sequence components of $I_a, I_b$ and $I_c$ respectively with phase sequence acb.

$I_{a0}, I_{b0}, I_{c0}$ = Zero sequence components of $I_a, I_b$ and $I_c$ respectively.

The vector diagram of positive, negative and zero sequence components are shown in fig 1.14.

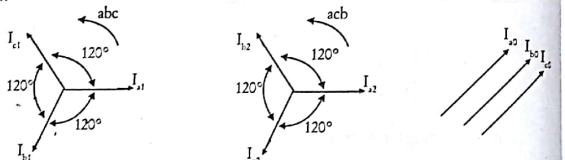



*Fig a : Positive sequence components* *Fig b : Negative sequence components* *Fig c : Zero sequence components*

*Fig 1.14 : Vector diagram of symmetrical components of unbalanced 3-phase current vectors*

The following equations are used to compute the unbalanced current vectors from the knowledge of their symmetrical components [Refer equations (1.92) to (1.95)].

$$I_a = I_{a0} + I_{a1} + I_{a2} \quad \text{.....(1.106)}$$

$$I_b = I_{a0} + a^2 I_{a1} + a I_{a2} \quad \text{.....(1.107)}$$

$$I_c = I_{a0} + a I_{a1} + a^2 I_{a2} \quad \text{.....(1.108)}$$

$$\begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} \quad \text{.....(1.109)}$$



The following equations are used to compute the symmetrical components of unbalanced current vectors [Refer equations (1.102) to (1.105)].

$$I_{a0} = \frac{1}{3}\left[I_a + I_b + I_c\right] \quad \text{.....(1.110)}$$

$$I_{a1} = \frac{1}{3}\left[I_a + aI_b + a^2I_c\right] \quad \text{.....(1.111)}$$

$$I_{a2} = \frac{1}{3}\left[I_a + a^2 I_b + a I_c\right] \quad \text{.....(1.112)}$$

$$\begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} \quad \text{.....(1.113)}$$

## EXAMPLE 1.19

The voltages across a 3 phase unbalanced load are $V_a = 300\angle 20°V$, $V_b = 360\angle 90°V$ and $V_c = 500\angle -140°V$. Determine the symmetrical components of voltages. Phase sequence is abc.

### SOLUTION

The symmetrical components of $V_a$ are given by the following matrix equations.

$$\begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix}$$

$$\therefore V_{a0} = \frac{1}{3}\left[V_a + V_b + V_c\right]$$

$$V_{a1} = \frac{1}{3}\left[V_a + aV_b + a^2V_c\right]$$

$$V_{a2} = \frac{1}{3}\left[V_a + a^2 V_b + a V_c\right]$$

Given that $V_a = 300\angle 20°V = 281.91 + j102.61\ V$

$V_b = 360\angle 90°V = 0 + j360\ V$

$V_c = 500\angle -140°V = -383.02 - j321.39\ V$

$\therefore aV_b = 1\angle 120° \times 360\angle 90° = 360\angle 210° = -311.77 - j180\ V$

$a^2V_b = 1\angle 240° \times 360\angle 90° = 360\angle 330° = 311.77 - j180\ V$

$aV_c = 1\angle 120° \times 500\angle -140° = 500\angle -20° = 469.85 - j171.01\ V$

$a^2V_c = 1\angle 240° \times 500\angle -140° = 500\angle 100° = -86.82 + j492.40\ V$

$$V_{a0} = \frac{1}{3}\left[V_a + V_b + V_c\right] = \frac{1}{3}(281.91 + j102.61 + 0 + j360 - 383.02 - j321.39)$$

$$= \frac{1}{3}(-101.11 + j141.22) = -33.70 + j47.07 = 57.89\angle 126°\ V$$

$$V_{a1} = \frac{1}{3}\left[V_a + a\,V_b + a^2\,V_c\right] = \frac{1}{3}(281.91 + j102.61 - 311.77 - j180 - 86.82 + j492.40)$$

$$= \frac{1}{3}(-116.68 + j415.01) = -38.89 + j138.34 = 143.70\angle 106°\ V$$

$$V_{a2} = \frac{1}{3}\left[V_a + a^2\,V_b + a\,V_c\right] = \frac{1}{3}(281.91 + j102.61 + 311.77 - j180 + 469.85 - j171.01)$$

$$= \frac{1}{3}(1063.53 - j248.40) = 354.51 - j82.80 = 364.05\angle -13°\ V$$

We know that $V_{a0} = V_{b0} = V_{c0}$

$\therefore$ The zero sequence components are

$V_{a0} = 57.89\angle 126°\ V$

$V_{b0} = 57.89\angle 126°\ V$

$V_{c0} = 57.89\angle 126°\ V$

We know that, $V_{b1} = a^2V_{a1}$ ; $V_{c1} = a\,V_{a1}$

$\therefore$ The positive sequence components are

$V_{a1} = 143.70\angle 106°\ V$

$V_{b1} = a^2V_{a1} = 1\angle 240° \times 143.70\angle 106° = 143.70\angle 346°\ V$

$V_{c1} = a\,V_{a1} = 1\angle 120° \times 143.70\angle 106° = 143.70\angle 226°\ V$

We know that, $V_{b2} = a\,V_{a2}$; $V_{c2} = a^2V_{a2}$

$\therefore$ The negative sequence components are

$V_{a2} = 364.05\angle -13°\ V$

$V_{b2} = a\,V_{a2} = 1\angle 120° \times 364.05\angle -13° = 364.05\angle 107°\ V$

$V_{c2} = a^2V_{a2} = 1\angle 240° \times 364.05\angle -13° = 364.05\angle 227°\ V$



## EXAMPLE 1.20

The symmetrical components of phase-a voltage in a 3-phase unbalanced system are $V_{a0} = 10\angle 180°V$, $V_{a1} = 50\angle 0°V$ and $V_{a2} = 20\angle 90°V$. Determine the phase voltages $V_a$, $V_b$ and $V_c$.

### SOLUTION

The phase voltages of $V_a$, $V_b$ and $V_c$ are given by the following matrix equations.

$$\begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix}$$

$\therefore V_a = V_{a0} + V_{a1} + V_{a2}$

$V_b = V_{a0} + a^2V_{a1} + a\,V_{a2}$

$V_c = V_{a0} + a\,V_{a1} + a^2V_{a2}$

Given that $V_{a0} = 10\angle 180°\ V = -10 + j0$

$V_{a1} = 50\angle 0°\ V = 50 + j0$

$V_{a2} = 20\angle 90°V = 0 + j20$

$\therefore aV_{a1} = 1\angle 120° \times 50\angle 0° = 50\angle 120° = -25 + j43.30$

$a^2V_{a1} = 1\angle 240° \times 50\angle 0° = 50\angle 240° = -25 - j43.30$

$aV_{a2} = 1\angle 120° \times 20\angle 90° = 20\angle 210° = -17.32 - j10$

$a^2V_{a2} = 1\angle 240° \times 20\angle 90° = 20\angle 330° = 17.32 - j10$

$V_a = V_{a0} + V_{a1} + V_{a2} = -10 + 50 + j20 = 40 + j20 = 44.72\angle 27°\ V$

$V_b = V_{a0} + a^2V_{a1} + aV_{a2} = -10 - 25 - j43.30 - 17.32 - j10$

$= -52.32 - j53.30 = 74.69\angle -134°\ V$

$V_c = V_{a0} + a\,V_{a1} + a^2\,V_{a2} = -10 - 25 + j43.30 + 17.32 - j10$

$= -17.68 + j33.3 = 37.70\angle 118°\ V$

## EXAMPLE 1.21

The symmetrical components of phase-a fault current in a 3-phase unbalanced system are $I_{a0} = 350\angle 90°$ A, $I_{a1} = 600\angle -90°$ A and $I_{a2} = 250\angle 90°$ A. Determine the phase currents $I_a$, $I_b$ and $I_c$.

### SOLUTION

The currents $I_a$, $I_b$ and $I_c$ are given by the following matrix equations.

$$\begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix}$$

$$\therefore I_a = I_{a0} + I_{a1} + I_{a2}$$

$$I_b = I_{a0} + a^2 I_{a1} + a I_{a2}$$

$$I_c = I_{a0} + a I_{a1} + a^2 I_{a2}$$

Given that $I_{a0} = 350\angle 90° = 0 + j350$

$$I_{a1} = 600\angle -90° = 0 - j600$$

$$I_{a2} = 250\angle 90° = 0 + j250$$

$$\therefore a I_{a1} = 1\angle 120° \times 600\angle -90° = 600\angle 30° = 519.2 + j300$$

$$a^2 I_{a1} = 1\angle 240° \times 600\angle -90° = 600\angle 150° = -519.62 + j300$$

$$a I_{a2} = 1\angle 120° \times 250\angle 90° = 250\angle 210° = -216.51 - j125$$

$$a^2 I_{a2} = 1\angle 240° \times 250\angle 90° = 250\angle 330° = 216.51 - j125$$

$$I_a = I_{a0} + I_{a1} + I_{a2} = j350 - j600 + j250 = 0$$

$$I_b = I_{a0} + a^2 I_{a1} + a I_{a2} = j350 - 519.62 + j300 - 216.51 - j125$$

$$= -736.13 + j525 = 904.16\angle 145° \text{ A}$$

$$I_c = I_{a0} + a I_{a1} + a^2 I_{a2} = j350 + 519.62 + j300 + 216.51 - j125$$

$$= 736.13 + j525 = 904.16\angle 35° \text{ A}$$



## EXAMPLE 1.22

Determine the symmetrical components of the unbalanced three phase currents $I_a = 10\angle 0°$ A, $I_b = 12\angle 230°$ A and $I_c = 10\angle 130°$ A.

### SOLUTION

The symmetrical components of $I_a$ given by the following matrix equations.

$$\begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix}$$

$$\therefore I_{a0} = \frac{1}{3}(I_a + I_b + I_c)$$

$$I_{a1} = \frac{1}{3}(I_a + a I_b + a^2 I_c)$$

$$I_{a2} = \frac{1}{3}(I_a + a^2 I_b + a I_c)$$

Given that $I_a = 10\angle 0° = 10 + j0$

$$I_b = 12\angle 230° = -7.71 - j9.19$$

$$I_c = 10\angle 130° = -6.43 + j7.66$$

$$\therefore a I_b = 1\angle 120° \times 12\angle 230° = 12\angle 350° = 11.82 - j2.08$$

$$a^2 I_b = 1\angle 240° \times 12\angle 230° = 12\angle 470° = 12\angle 110° = -4.10 + j11.28$$

$$a I_c = 1\angle 120° \times 10\angle 130° = 10\angle 250° = -3.42 - j9.40$$

$$a^2 I_c = 1\angle 240° \times 10\angle 130° = 10\angle 370° = 10\angle 10° = 9.85 + j1.74$$

$$I_{a0} = \frac{1}{3}(I_a + I_b + I_c) = \frac{1}{3}(10 - 7.71 - j9.19 - 6.43 + j7.66)$$

$$= \frac{1}{3}(-4.14 - j1.53) = -1.38 - j0.51 = 1.47\angle -160°$$

$$I_{a1} = \frac{1}{3}(I_a + a I_b + a^2 I_c) = \frac{1}{3}(10 + 11.82 - j2.08 + 9.85 + j1.74)$$

$$= \frac{1}{3}(31.67 - j0.34) = 10.56 - j0.11 = 10.56\angle -0.6° \approx 10.56\angle 0°$$

$$I_{a2} = \frac{1}{3}(I_a + a^2 I_b + a I_c) = \frac{1}{3}(10 - 4.10 + j11.28 - 3.42 - j9.40)$$

$$= \frac{1}{3}(2.48 + j1.88) = 0.83 + j0.63 = 1.04\angle 37°$$

We know that $I_{a0} = I_{b0} = I_{c0}$

∴ The zero sequence components are

$I_{a0} = 1.47\angle -160°$ A

$I_{b0} = 1.47\angle -160°$ A

$I_{c0} = 1.47\angle -160°$ A.

We know that, $I_{b1} = a^2 I_{a1}$ ; $I_{c1} = a I_{a1}$

∴ The positive sequence components are

$I_{a1} = 10.56\angle 0°$ A

$I_{b1} = a^2 I_{a1} = 1\angle 240° \times 10.56\angle 0° = 10.56\angle 240°$ A

$I_{c1} = a I_{a1} = 1\angle 120° \times 10.56\angle 0° = 10.56\angle 120°$ A

We know that, $I_{b2} = a I_{a2}$ ; $I_{c2} = a^2 I_{a2}$

∴ The negative sequence components are

$I_{a2} = 1.04\angle 37°$ A

$I_{b2} = a I_{a2} = 1\angle 120° \times 1.04\angle 37° = 1.04\angle 157°$ A

$I_{c2} = a^2 I_{a2} = 1\angle 240° \times 1.04\angle 37° = 1.04\angle 277°$ A

## 1.8 SEQUENCE IMPEDANCES AND SEQUENCE NETWORKS

The sequence impedances are impedances offered by the circuit elements (or power system components) to positive, negative and zero sequence currents. In any element of a circuit, the voltage drop caused by current of a certain sequence depends on the impedance of the element to that sequence current.

The impedance of a circuit element when positive sequence currents alone are flowing is called the positive sequence impedance. Similarly, when only negative sequence currents are present, the impedance is called negative sequence impedance. When only zero sequence currents are present the impedance is called zero sequence impedance. The impedance of any element of a balanced circuit to current of one sequence may be different from impedance to current of another sequence.

The single phase equivalent circuit of power system (impedance or reactance diagram) formed using the impedances of any one sequence only is called the sequence network for that particular sequence. Therefore the impedance or reactance diagram formed using positive sequence impedance is called positive sequence network. Similarly the impedance or reactance diagram formed using negative sequence impedance is called negative sequence network. The impedance or reactance diagram formed using zero sequence impedance is called zero sequence network.



The sequence impedances and networks are useful in the analysis of unsymmetrical faults in the power system. In unsymmetrical fault analysis of a power system, the positive, negative and zero sequence networks of the system are determined and then they are interconnected to represent the various unbalanced fault conditions. Each sequence network includes the generated emfs and impedances of like sequence. Also, the sequence network carries only the current of like sequence.

### Sequence impedances and networks of generator

Consider the three phase equivalent circuit of a generator shown in fig 1.15. The neutral of the generator is grounded through a reactance, $Z_n$.

When the generator is delivering a balanced load or under symmetrical fault the neutral current is zero. But when the generator is delivering an unbalanced load or during unsymmetrical faults the neutral current flows through $Z_n$.

The generator is designed to supply balanced three phase voltages. Therefore the generated emfs are of positive sequence only.

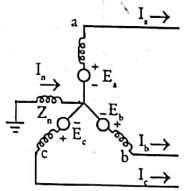



*Fig 1.15 : Three phase equivalent circuit of a generator grounded through a reactance*

Let $E_a, E_b, E_c$ = Generated emf per phase in phase a, b and c respectively. (Positive sequence emf)

$Z_1$ = Positive sequence impedance per phase of generator.

$Z_2$ = Negative sequence impedance per phase of generator.

$Z_{g0}$ = Zero sequence impedance per phase of generator.

$Z_n$ = Neutral reactance.

$Z_0$ = Total zero sequence impedance per phase of zero-sequence network of generator.

The positive sequence network consists of an emf in series with positive sequence impedance of the generator. The negative and zero sequence network will not have any sources but include their respective sequence impedance. The positive, negative and zero sequence current paths are shown in fig 1.16. The positive, negative and zero sequence networks of the generator are shown in fig 1.17.

*Note : The positive and negative sequence currents are balanced currents and so they will not pass through neutral reactance.*

The reactances in positive sequence network is subtransient, transient or synchronous reactance depending on whether subtransient, transient or steady state conditions are being studied. Under no load condition the emf $E_a$ is the induced emf per phase. Under load or fault condition $E_a$ is replaced by $E_g'$ for transient state and $E_a$ is replaced by $E_g''$ for subtransient state.

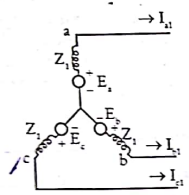


*Fig (a) : Positive-sequence current paths.*

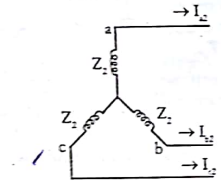


*Fig (b) : Negative-sequence current paths.*

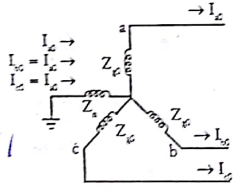


*Fig (c) : Zero-sequence current paths.*

*Fig 1.16 : Sequence current paths in a generator*

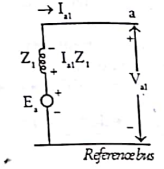


*Fig (a) : Positive-sequence network*

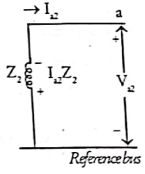


*Fig (b) : Negative-sequence network*

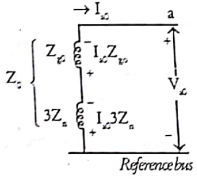


*Fig (c) : Zero-sequence network*

*Fig 1.17 : Sequence networks of a generator*



On examining the zero sequence current paths (refer fig 1.16-c) we can say that the current through neutral reactance is $3I_{a0}$. The zero-sequence voltage drop from point-a to ground is $-3I_{a0}Z_n - I_{a0}Z_{g0}$.

The zero sequence network is a single phase network and assumed to carry only the zero sequence current of one phase. Hence the zero sequence current of one phase, must have an impedance of $3Z_n + Z_{g0}$

$$\left.\begin{array}{l}\therefore \text{Total zero sequence impedance per phase} \\ \text{of a generator grounded through reactance}\end{array}\right\} Z_0 = 3Z_n + Z_{g0} \qquad \text{.....(1.114)}$$

With reference to fig 1.17, the equations for the phase-a component voltages are,

$$V_{a1} = E_a - I_{a1}Z_1 \qquad \text{.....(1.115)}$$

$$V_{a2} = -I_{a2}Z_2 \qquad \text{.....(1.116)}$$

$$V_{a3} = -I_{a0}Z_0 \qquad \text{.....(1.117)}$$

The zero sequence network of generator when the neutral is solidly grounded (i.e. directly grounded) and when the neutral is ungrounded are shown in fig 1.18 and fig 1.19 respectively. In these cases there is no change in positive and negative sequence network.

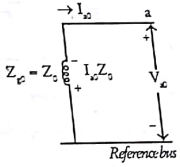


*Fig 1.18 : Zero-sequence network of a generator when the neutral is solidly grounded*

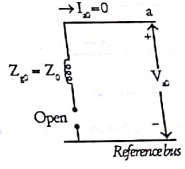


*Fig 1.19 : Zero-sequence network of a generator when the neutral is ungrounded*

**Note** : The sequence networks of synchronous motor is same as that of generator when the directions of currents in the sequence network of generator are reversed.

## Sequence impedances and networks of transmission lines

The impedance per phase of transmission line for balanced currents is independent of phase sequence. This is due to the symmetry of transposed transmission lines. Therefore, the impedances offered by the transposed transmission lines for positive and negative sequence currents are identical.

The zero sequence current is identical (both in magnitude and phase) in each phase conductor and returns through the ground, through overhead ground wires or through both. The ground wires being grounded at several towers, the return currents in the ground wire may not be uniform along the entire length of transmission line. But for positive or negative sequence currents there is no return current and they have a phase difference of 120°. Therefore the magnetic field due to zero sequence current is different from the magnetic field caused by either positive or negative sequence current. Due to the difference in the magnetic field, the zero sequence inductive reactance is 2 to 3.5. times the positive sequence reactance.

Let, $Z_1$ = Positive sequence impedance of transmission line
$Z_2$ = Negative sequence impedance of transmission line
$Z_0$ = Zero sequence impedance of transmission line

The positive, negative and zero sequence impedances of transmission lines are represented as a series impedance in their respective sequence networks as shown in fig 1.20.

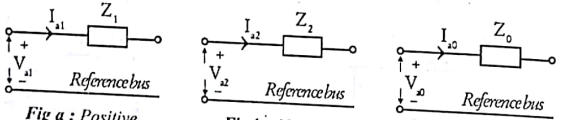



*Fig a : Positive sequence network* *Fig b : Negative sequence network* *Fig c : Zero sequence network*

*Fig 1.20 : Positive, Negative and zero sequence networks of transmission line*

## Sequence impedances and networks of transformer

When the applied voltage is balanced, the positive and negative sequence of linear, symmetrical, static devices are identical. Therefore in a transformer the positive and negative sequence impedances are identical. Even though the zero sequence impedance may slightly differ from positive and negative sequence impedance, it is normal practice to assume the zero sequence impedance as equal to positive or negative sequence impedance. [For all types of transformers the series impedances of all sequences are assumed equal ].

**Note** : *When the neutral of star connection is grounded through reactance $Z_n$ then $3Z_n$ should be added to zero sequence impedance of transformer to get the total zero sequence impedance.*



Let, $Z_1$ = Positive sequence impedance of transformer
$Z_2$ = Negative sequence impedance of transformer
$Z_0$ = Zero sequence impedance of transformer

The positive and negative sequence impedances of transformer are represented as a series impedance in their respective sequence networks as shown in fig 1.21.

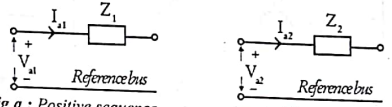



*Fig a : Positive sequence network* *Fig b : Negative sequence network*

*Fig 1.21 : Positive and Negative sequence networks of transformer*

The zero-sequence network of the transformer depends on the type of connections (Y or Δ) of the primary and secondary windings and also on the grounding of neutral in Y connection.

The following general observations can be made for zero sequence currents in transformers.

1. When magnetizing current is neglected, the primary winding will carry current only if there is a current flow on the secondary winding. Therefore the zero sequence current can flow in the primary winding of a transformer only if there is a path for zero sequence current in secondary winding or vice-versa.
2. If the neutral point in the Y connected winding is not grounded then there is no path for zero sequence current in star connected winding.
3. The zero sequence current flows in the star connected winding and in the lines connected to the winding only when the neutral point is grounded.
4. The zero sequence current can circulate in the delta connected winding but the zero sequence current cannot flow through the lines connected to the delta connected winding.

Based on the above observations the zero sequence network of 3-phase transformer can be obtained for any configuration. The zero sequence network for nine possible configuration are presented in table-1.3. The arrows on the windings, indicate path for zero sequence current and the absence of arrows indicate that there is no path for zero sequence currents.

**Table-1.3 : Zero sequence network of three phase transformer**

| Configuration | Winding Connection Diagram | Zero sequence Network |
|---|---|---|
| M N<br>Y Y | M N | M $Z_0$ N<br>Reference bus |
| M N<br>$Y_n$ Y | M N | M $Z_0$ N<br>Reference bus |
| M N<br>Y $Y_n$ | M N | M $Z_0$ N<br>Reference bus |
| M N<br>$Y_n$ $Y_n$ | M N | M $Z_0$ N<br>Reference bus |
| M N<br>Y Δ | M N | M $Z_0$ N<br>Reference bus |
| M N<br>Δ Y | M N | M $Z_0$ N<br>Reference bus |
| M N<br>Δ | M N | M $Z_0$ N<br>Reference bus |
| M N<br>Δ $Y_n$ | M N | M $Z_0$ N<br>Reference bus |
| M N<br>Δ Δ | M N | M $Z_0$ N<br>Reference bus |



## Sequence impedances and networks of loads

In balanced Y or Δ connected loads the positive, negative and zero sequence impedances are equal. When the neutral point of star connected load is grounded through a reactance $Z_n$ then $3Z_n$ is added to the zero sequence impedance of load to get the total zero sequence impedance of load.

Let $Z_{L1}$ = Positive sequence impedance of load
$Z_{L2}$ = Negative sequence impedance of load.
$Z_{L0}$ = Zero sequence impedance of load.

The positive and negative sequence impedances of load are represented as a shunt impedance in their respective sequence network as shown in fig 1.22.

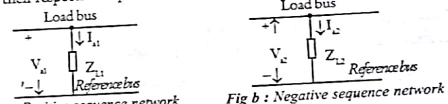



*Fig a : Positive sequence network* *Fig b : Negative sequence network*

***Fig 1.22 : Positive and Negative sequence network of load***

The zero sequence network of the 3-phase load depends on the type of connection, i.e., Y or Δ connection. The zero sequence current will flow in network only if a return path exists for it. The zero sequence network for various types of loads are shown in table-1.4.

**Table-1.4 : Zero sequence networks of loads**

| Connection Diagram of load | Zero sequence network |
|---|---|
| | Load bus<br>$Z_{L0}$<br>Reference bus |
| | Load bus<br>$Z_{L0}$<br>Reference bus |
| $Z_n$ | Load bus<br>$Z_{L0}$<br>$3Z_n$<br>Reference bus |
| | Load bus<br>$Z_{L0}$<br>Reference bus |

## EXAMPLE 1.23

Draw the positive, negative and zero sequence reactance diagram of the power system shown in fig 1.23.1.

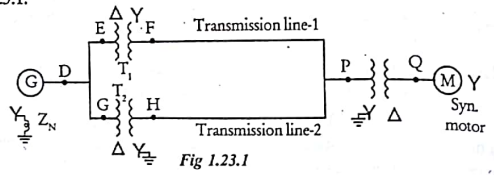



*Fig 1.23.1*

### SOLUTION

The positive, negative and zero sequence reactance diagrams (networks) of the power system are shown in fig (1.23.2), (1.23.3) and (1.23.4) respectively.

Let, $X_{G,1}$ = Positive sequence reactance of generator G
$X_{M,1}$ = Positive sequence reactance of motor M
$X_{T1,1}$ = Positive sequence reactance of transformer T1
$X_{T2,1}$ = Positive sequence reactance of transformer T2
$X_{T3,1}$ = Positive sequence reactance of transformer T3
$X_{TL,1}$ = Positive sequence reactance of transmission line-1
$X_{TL,2}$ = Positive sequence reactance of transmission line-2

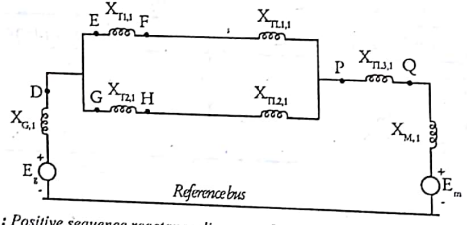
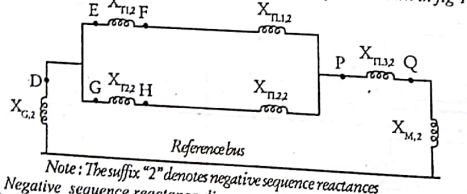


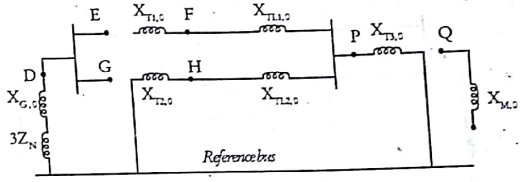

*Fig 1.23.2 : Positive sequence reactance diagram of power system shown in fig 1.23.1*




*Note : The suffix "2" denotes negative sequence reactances*

*Fig 1.23.3 : Negative sequence reactance diagram of power system shown in fig 1.23.1*






*Note : The suffix "0" denotes zero sequence reactances*

*Fig 1.23.4 : Zero sequence reactance diagram of power system shown in fig 1.23.1*

## EXAMPLE 1.24

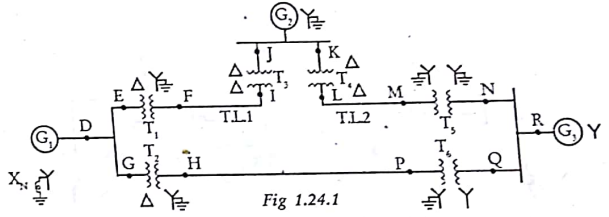



*Fig 1.24.1*

For the power system shown in fig 1.24.1, draw the positive, negative and zero sequence reactance diagrams.

### SOLUTION

The positive, negative and zero sequence reactance diagrams (networks) of the power system are shown in fig (1.24.2), (1.24.3) and (1.24.4) respectively.

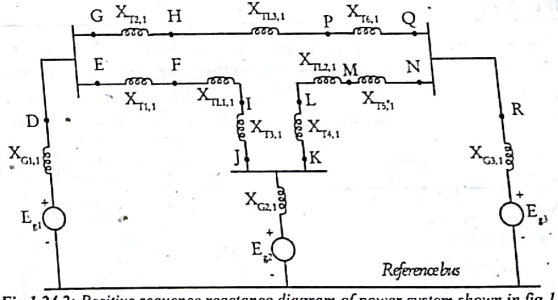



*Fig 1.24.2: Positive sequence reactance diagram of power system shown in fig 1.24.1*

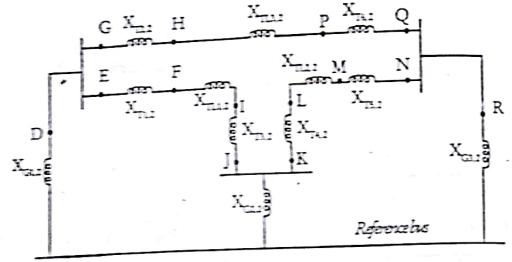



*Fig 1.24.3: Negative sequence reactance diagram of power system shown in fig 1.24.1*

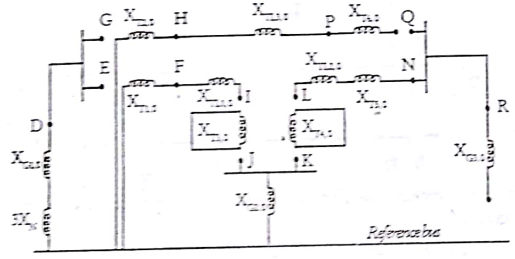



*Fig 1.24.4: Zero sequence reactance diagram of power system shown in fig 1.24.1*

## EXAMPLE 1.25

Determine the positive, negative and zero sequence networks for the system shown in fig 1.25.1. Assume zero sequence reactances for the generator and synchronous motors as 0.06 p.u. Current limiting reactors of 2.5 Ω are connected in the neutral of the generator and motor No.2. The zero sequence reactance of the transmission line is j300Ω.

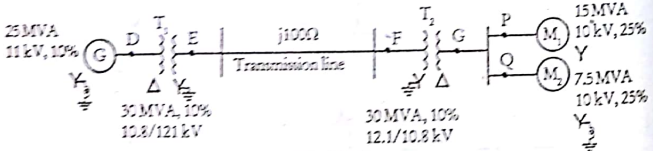



*Fig 1.25.1*



### SOLUTION

Let us choose the generator ratings as new base values for entire system.

Base megavoltampere, $MVA_{b,new} = 25\,MVA$

Base kilovolt, $kV_{b,new} = 11\,kV$

### Sequence reactances of Generator G

Since the generator rating and the new base values are same, the generator p.u. reactances does not change. Also for generator the positive and negative sequence reactances are same.

∴Positive sequence reactance of generator, $X_{G,1} = 10\% = 10/100 = 0.1$ p.u.

Negative sequence reactance of generator, $X_{G,2} = 0.1$ p.u.

Zero sequence reactance of generator, $X_{G,0} = 0.06$ p.u.

$$\text{Base impedance, } Z_b = \frac{(kV_{b,new})^2}{MVA_{b,new}} = \frac{11^2}{25} = 4.84\Omega$$

$$\left.\begin{matrix}\text{p.u. value of generator}\\ \text{neutral reactance}\end{matrix}\right\} X_{GN} = \frac{\text{Actual Neutral reactance}}{\text{Base impedance}} = \frac{2.5}{4.84} = 0.517 \text{ p.u.}$$

### Sequence reactances of Transformer $T_1$

$$\left.\begin{matrix}\text{New p.u. reactance}\\ \text{of transformer } T_1\end{matrix}\right\} = X_{pu,old} \times \left(\frac{kV_{b,old}}{kV_{b,new}}\right)^2 \times \frac{MVA_{b,new}}{MVA_{b,old}}$$

Here, $X_{pu,old} = 10\% = 0.1$, $kV_{b,old} = 10.8$ kV, $MVA_{b,old} = 30$ MVA

$kV_{b,new} = 11$ kV, $MVA_{b,new} = 25$ MVA

$$\left.\begin{matrix}\text{New p.u. reactance}\\ \text{of transformer } T_1\end{matrix}\right\} = 0.1 \times \left(\frac{10.8}{11}\right)^2 \times \left(\frac{25}{30}\right) = 0.08 \text{ p.u.}$$

In transformer the specified reactance is positive sequence reactance. Also we assume that the positive, negative and zero sequence reactances of the transformer are equal.

∴Positive sequence reactance of transformer $T_1$, $X_{T1,1} = 0.08$ p.u.

Negative sequence reactance of transformer $T_1$, $X_{T1,2} = 0.08$ p.u.

Zero sequence reactance of transformer $T_1$, $X_{T1,0} = 0.08$ p.u.

### Sequence reactances of Transmission line

$$\left.\begin{matrix}\text{The base kV on HT side}\\ \text{of transformer } T_1\end{matrix}\right\} = \text{Base kV on LT side} \times \frac{\text{HT voltage rating}}{\text{LT voltage rating}}$$

$$= 11 \times \frac{121}{10.8} = 123.24 \text{ kV}$$

Now, $kV_{b,new} = 123.24$ kV

$$\text{Base impedance, } Z_b = \frac{(kV_{b,new})^2}{MVA_{b,new}} = \frac{(123.24)^2}{30} = 506.27\Omega$$

$$\left.\begin{matrix}\text{p.u. reactance of}\\ \text{transmission line}\end{matrix}\right\} = \frac{\text{Actual reactance}}{\text{Base impedance}} = \frac{100}{506.27} = 0.198 \text{ p.u.}$$

The specified reactance in single line diagram is positive sequence reactance. Also the negative sequence reactance of a transmission line is same as that of positive sequence reactance.

$\therefore$ Positive sequence reactance of transmission line, $X_{TL,1} = 0.198$ p.u

Negative sequence reactance of transmission line, $X_{TL,2} = 0.198$ p.u

$$\left.\begin{matrix}\text{p.u. value of zero sequence}\\ \text{reactance of transmission line}\end{matrix}\right\} X_{TL,0} = \frac{\text{Zero sequence reactance in } \Omega}{\text{Base impedance}} = \frac{300}{506.27} = 0.593 \text{ p.u.}$$

### Sequence reactances of Transformer $T_2$

The ratings and winding connections of transformer $T_1$ and $T_2$ are identical and so the sequence reactances of $T_1$ and $T_2$ are same.

Positive sequence reactance of transformer $T_2$, $X_{T2,1} = 0.08$ p.u.

Negative sequence reactance of transformer $T_2$, $X_{T2,2} = 0.08$ p.u.

Zero sequence reactance of transformer $T_2$, $X_{T2,0} = 0.08$ p.u.

### Sequence reactances of Synchronous motor $M_1$

$$\left.\begin{matrix}\text{Base kV on LT side}\\ \text{of transformer } T_2\end{matrix}\right\} = \text{Base kV on HT side} \times \frac{\text{LT voltage rating}}{\text{HT voltage rating}}$$

$$= 123.24 \times \frac{10.8}{121} = 11 \text{ kV}$$

Now, $kV_{b,new} = 11$ kV

$$\left.\begin{matrix}\text{New p.u. reactance}\\ \text{of motor } M_1\end{matrix}\right\} = X_{pu,old} \times \left(\frac{kV_{b,old}}{kV_{b,new}}\right)^2 \times \frac{MVA_{b,new}}{MVA_{b,old}}$$



Here, $X_{pu,old} = 25\% = 0.25$ ; $kV_{b,old} = 10$ kV ; $MVA_{b,old} = 15$ MVA

$kV_{b,new} = 11$ kV ; $MVA_{b,new} = 25$ MVA

$$\left.\begin{matrix}\text{New p.u. reactance}\\ \text{of motor } M_1\end{matrix}\right\} = 0.25 \times \left(\frac{10}{11}\right)^2 \times \frac{25}{15} = 0.344 \text{ p.u.}$$

The reactance specified in single line diagram is positive sequence reactance. Also the negative sequence reactance of synchronous motor is same as that of positive sequence reactance.

$\therefore$ Positive sequence reactance of motor $M_1$, $X_{M1,1} = 0.344$ p.u.

Negative sequence reactance of motor $M_1$, $X_{M1,2} = 0.344$ p.u.

$$\left.\begin{matrix}\text{The zero sequence reactance}\\ \text{of motor } M_1 \text{ on new bases}\end{matrix}\right\} X_{M1,0} = X_{pu,old} \times \left(\frac{kV_{b,old}}{kV_{b,new}}\right)^2 \times \frac{MVA_{b,new}}{MVA_{b,old}}$$

$$= 0.06 \times \left(\frac{10}{11}\right)^2 \times \frac{25}{15} = 0.083 \text{ p.u.}$$

### Sequence reactances of Synchronous motor $M_2$

$$\left.\begin{matrix}\text{New p.u. reactance}\\ \text{of motor } M_2\end{matrix}\right\} = X_{pu,old} \times \left(\frac{kV_{b,old}}{kV_{b,new}}\right)^2 \times \frac{MVA_{b,new}}{MVA_{b,old}}$$

Here, $X_{pu,old} = 25\% = 0.25$ ; $kV_{b,old} = 10$ kV ; $MVA_{b,old} = 7.5$ MVA

$kV_{b,new} = 11$ kV ; $MVA_{b,new} = 25$ MVA

$$\left.\begin{matrix}\text{New p.u. reactance}\\ \text{of motor } M_2\end{matrix}\right\} = 0.25 \times \left(\frac{10}{11}\right)^2 \times \frac{25}{7.5} = 0.689 \text{ p.u.}$$

The reactance specified in single line diagram is positive sequence reactance. Also the negative sequence reactance of synchronous motor is same as that of positive sequence reactance.

$\therefore$ Positive sequence reactance of motor $M_2$, $X_{M2,1} = 0.689$ p.u.

Negative sequence reactance of motor $M_2$, $X_{M2,2} = 0.689$ p.u.

The zero sequence reactance of motor $M_1$ on the new bases:

$$X_{M2,0} = X_{pu,old} \times \left(\frac{kV_{b,old}}{kV_{b,new}}\right)^2 \times \frac{MVA_{b,new}}{MVA_{b,old}}$$

$$= 0.06 \times \left(\frac{10}{11}\right)^2 \times \frac{25}{7.5} = 0.165 \text{ p.u.}$$

Base impedance, $Z_b = \frac{(kV_{b,new})^2}{MVA_{b,new}} = \frac{11^2}{25} = 4.84\Omega$

p.u. value of motor neutral reactance $X_{MN} = \frac{\text{Actual neutral reactance}}{\text{Base impedance}} = \frac{2.5}{4.84} = 0.517$ p.u.

### Positive sequence network

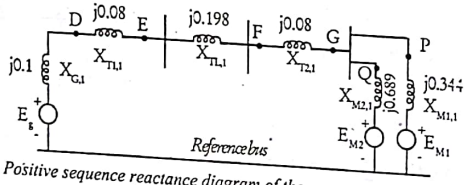



*Fig 1.25.2 : Positive sequence reactance diagram of the power system shown in fig 1.25.1*

### Negative sequence network

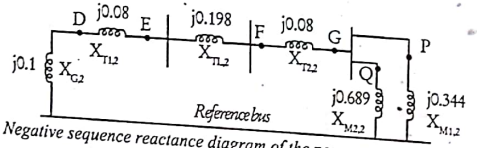



*Fig 1.25.3 : Negative sequence reactance diagram of the power system shown in fig 1.25.1*

### Zero sequence network

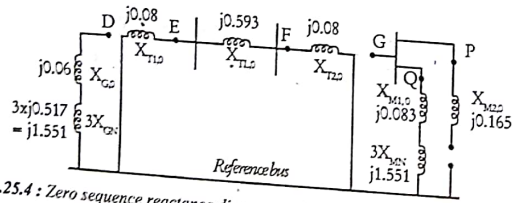



*Fig 1.25.4 : Zero sequence reactance diagram of the power system shown in fig 1.25.1*



## 1.9 SHORT QUESTIONS AND ANSWERS

1. *What is single line diagram ?*

A single line diagram is diagrammatic representation of power system in which the components are represented by their symbols and the interconnection between them are shown by a single straight line (eventhough the system is 3-phase system). The ratings and the impedances of the components are also marked on the single line diagram.

2. *What are the components of power system?*

The components of power system are Generators, Power Transformers, Transmission lines, Substation Transformers, Distribution Transformers and Loads.

3. *Draw the symbols used to represent various components in a power system.*

The symbols used to represent various components are shown in the following table.

| | | | |
|---|---|---|---|
| Machine or rotating armature | ◯ | Power circuit breaker, (oil/gas filled) | —□— |
| Two-winding power transformer | ⧘ | Air circuit breaker | —⌒— |
| Three-winding power transformer | ⧘ | Three-phase, three-wire delta connection | △ |
| Fuse | —▭— | Three-phase star, neutral ungrounded | Y |
| Current transformer | —∩— | Three-phase star, neutral grounded | Y⏚ |
| Potential transformer | —)( or —)( | Ammeter and voltmeter | —(A)— —(V)— |

4. *Define per unit value.*

The per unit value of any quantity is defined as the ratio of the actual value of the quantity to the base value expressed as a decimal. The base value is an arbitrary chosen value of the quantity.

$$\text{Per unit value} = \frac{\text{Actual value}}{\text{Base value}}$$

5. *What are the quantities whose base values are required to represent the power system by reactance diagram ?*

The base values of Voltage, Current, Power and Impedance are required to represent the power system by reactance diagram. Selection of base values for any two of them determines the base values of the remaining two. Usually the base values of voltage and power are chosen in kilovolt and kVA or MVA respectively. The base values of current and impedance are calculated using the chosen bases.

6. *What is the need for base values?*

The components or various sections of power system may operate at different voltage and power levels. It will be convenient for analysis of power system if the voltage, power, current and impedance ratings of components of power system are expressed with reference to a common value called base value. Hence for analysis purpose a base value is chosen for voltage, power, current and impedance. Then all the voltage, power, current and impedance ratings of the components are expressed as a percent or per unit of the base value.

7. *Write the equation for converting the p.u. impedance expressed in one base to another?*

$$Z_{pu,new} = Z_{pu,old} \times \left(\frac{kV_{b,old}}{kV_{b,new}}\right)^2 \times \frac{MVA_{b,new}}{MVA_{b,old}}$$

8. *What are the advantages of per-unit computations?*

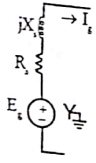
(i) Manufacturers usually specify the impedance of a device or machine in per unit on the base of the name plate rating.

(ii) The p.u. values of widely different rating machines lie within a narrow range, eventhough the ohmic values has a very large range.

(iii) The p.u. impedance of circuit element connected by transformers expressed on a proper base will be same if it is referred to either side of a transformer.

(iv) The p.u. impedance of a 3-phase transformer is independent of the type of winding connection (Y or Δ).

9. *Draw the equivalent circuit of a 3-phase generator.*




$X_s$ = Synchronous reactance per phase

$R_a$ = Armature resistance per phase

$E_g$ = Inducted emf per phase

$I_g$ = Current per phase.



10. *How the loads are represented in reactance or impedance diagram?*

The resistive and reactive loads can be represented by any one of the following representation.

(i) Constant power representation

Load power, $S = P + jQ$

(ii) Constant current representation

Load current, $I = \frac{\sqrt{P^2 + Q^2}}{|V|} \angle \delta - \theta$

(iii) Constant impedance representation

Load current, $Z = \frac{|V|^2}{P - jQ}$

11. *Draw the single phase equivalent circuit of a three-winding transformer.*

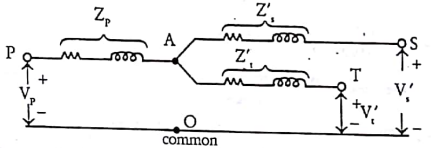



*Fig Q.11 : Single phase equivalent circuit of three winding transformer*

12. *A Y-Δ, 3-phase transformer bank is formed using three numbers of 1-phase transformers each rated at 300 kVA, 127/13.2 kV. What is the kVA and kV rating of the 3-phase bank?*

**Solution**

| | |
|---|---|
| kVA rating of 3-phase transformer | $= 3 \times 300 = 900$ kVA |
| Line voltage of Y-connected winding | $= \sqrt{3} \times 127 = 220$ kV |
| Line voltage of Δ-connected winding | $= 13.2$ kV |
| ∴The ratio of line voltages of 3-phase transformer bank | $= 220/13.2$ kV (Y/Δ) |

13. *If the reactance in ohms is 15 ohms, find the p.u. value for a base of 15 kVA and 10 kV.*

**Solution**

$$\text{Base impedance, } Z_b = \frac{(kV)^2}{MVA} = \frac{(kV)^2}{kVA/1000} = \frac{10^2}{15/1000} = 6666.67\Omega$$

$$\text{p.u. value of reactance} = \frac{\text{reactance in ohms}}{\text{Base impedance}} = \frac{15}{6666.67} = 0.0022 \text{ p.u.}$$

14. *A generator rated at 30 MVA, 11 kV has a reactance of 20%. Calculate its p.u. reactances for a base of 50 MVA and 10 kV.*

**Solution**

$$\text{New p.u. reactance of generator} = X_{pu,old} \times \left(\frac{kV_{b,old}}{kV_{b,new}}\right)^2 \times \frac{MVA_{b,new}}{MVA_{b,old}}$$

Here, $X_{pu,old} = 20\% = 0.2$ p.u. ; $kV_{b,old} = 11$ kV, $MVA_{b,old} = 30$ MVA

$kV_{b,new} = 10$ kV ; $MVA_{b,new} = 50$ MVA

$$\therefore \text{New p.u. reactance of generator} = 0.2 \times \left(\frac{11}{10}\right)^2 \times \frac{50}{30} = 0.403 \text{ p.u.}$$

15. *A Y-connected generator rated at 300 MVA, 33 kV, has a reactance of 1.24 p.u. Find the ohmic value of reactance.*

**Solution**

$$\text{Base impedance, } Z_b = \frac{(kV)^2}{MVA} = \frac{33^2}{300} = 3.63\ \Omega/\text{phase}$$

$$\text{We know that, p.u. reactance} = \frac{\text{Actual reactance}}{\text{Base impedance}}$$

$\therefore$ Reactance of generator = p.u. reactance $\times Z_b = 1.24 \times 3.63 = 4.5012\ \Omega$/phase

16. *The base kV and base MVA of a 3-phase transmission line is 33 kV and 10 MVA respectively. Calculate the base current and base impedance.*

**Solution**

$$\text{Base current, } I_b = \frac{kVA_b}{\sqrt{3}\,kV_b} = \frac{MVA_b \times 1000}{\sqrt{3}kV_b} = \frac{10 \times 1000}{\sqrt{3} \times 33} = 174.95A$$

$$\text{Base impedance, } Z_b = \frac{(kV_b)^2}{MVA_b} = \frac{33^2}{10} = 108.9\ \Omega/\text{phase}$$

17. *How the induction motor is represented in reactance diagram?*

For estimation of steady state fault current, the induction motor is neglected. But for estimation of fault current immediately after the fault (i.e., to estimate sub-transient fault currents), the induction motor can be represented by a source in series with reactance. The value of the source is the induced emf per phase and the value of reactance is total reactance of induction motor per phase referred to stator.

18. *What is impedance and reactance diagram?*

The impedance diagram is the equivalent circuit of power system in which the various components of power system are represented by their approximate or simplified equivalent circuits. The impedance diagram is used for load flow studies.



The reactance diagram is the simplified equivalent circuit of power system in which the various components are represented by their reactances. The reactance diagram can be obtained from impedance diagram if all the resistive components are neglected. The reactance diagram is used for fault calculations.

19. *What are the approximations made in impedance diagram?*

The following approximations are made while forming impedance diagram.

1. The neutral reactances are neglected.
2. The shunt branches in equivalent circuit of induction motor are neglected.

20. *What are the factors that need to be omitted for an impedance diagram to reduce it to a reactance diagram ? (or)*

*What are the approximations made in reactance diagram?*

The following approximations are made in reactance diagram

1. The neutral reactances are neglected.
2. Shunt branches in the equivalent circuits of transformer are neglected.
3. The resistances are neglected.
4. All static loads and induction motors are neglected.
5. The capacitance of the transmission lines are neglected.

21. *Give equations for transforming base kV on LV side to HV side of transformer and vice-versa.*

$$\text{Base kV on HT side} = \text{Base kV on LT side} \times \frac{\text{HT voltage rating}}{\text{LT voltage rating}}$$

$$\text{Base kV on LT side} = \text{Base kV on HT side} \times \frac{\text{LT voltage rating}}{\text{HT voltage rating}}$$

22. *For the power system shown in fig Q22, by taking generator rating as base values, specify the base values of the transmission line and motor circuit.*

The transmission line and motor circuit

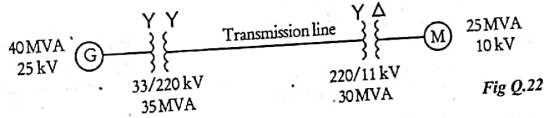



*Fig Q.22*

**Solution**

Generator circuit (selected base)

Base kilovolt = 25 kV

Base megavoltampere = 40 MVA

| Transmission line | Motor circuit |
|---|---|
| Base kilovolt = $25 \times \frac{220}{33} = 166.67$ kV | Base kilovolt = $166.67 \times \frac{11}{220} = 8.33$ kV |
| Base megavoltampere = 40 MVA | Base megavoltampere = 40 MVA |

23. *What is a bus?*

The meeting point of various components in a power system is called a bus. The bus is a conductor made of copper or aluminium having negligible resistance. The buses are considered as points of constant voltage in a power system.

24. *What is bus admittance matrix?*

The matrix consisting of the self and mutual admittances of the network of a power system is called bus admittance matrix. It is given by the admittance matrix **Y** in the node basis matrix equation of a power system and it is denoted as $\mathbf{Y}_{bus}$. The bus admittance matrix is symmetrical.

25. *Name the diagonal and off-diagonal elements of bus admittance matrix.*

The diagonal elements of bus admittance matrix are called self admittances of the buses and off-diagonal elements are called mutual admittances of the buses.

26. *Write the equation to find the elements of new bus admittance matrix after eliminating $n^{th}$ row and column in a n× n bus admittance matrix.*

The element $Y_{jk}$ of new bus admittance matrix is given by

$$Y_{jk,new} = Y_{jk} - \frac{Y_{jn}\, Y_{nk}}{Y_{nn}} \quad \text{for } j = 1, 2, 3, \ldots, (n-1) \text{ and } k = 1, 2, 3, \ldots, (n-1)$$

where $Y_{jk}$, $Y_{jn}$, $Y_{nk}$ and $Y_{nn}$ are elements of original or given bus admittance matrix of order $(n \times n)$.

27. *Form the bus admittance matrix for the system whose reactance diagram is shown in fig Q27.*

**Solution**

$$Y_{bus} = \begin{bmatrix} \frac{1}{j0.1} + \frac{1}{j0.25} + \frac{1}{j0.5} & -\left(\frac{1}{j0.25} + \frac{1}{j0.5}\right) \\ -\left(\frac{1}{j0.25} + \frac{1}{j0.5}\right) & \frac{1}{j0.25} + \frac{1}{j0.5} + \frac{1}{j0.2} \end{bmatrix}$$

$$= \begin{bmatrix} -j16 & j6 \\ j6 & -j11 \end{bmatrix}$$

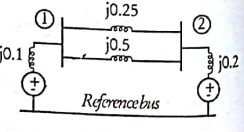



*Fig Q27*



28. *What is bus impedance matrix?*

The matrix consisting of driving point impedances and transfer impedances of the network of a power system is called bus impedance matrix. It is given by the inverse of bus admittance matrix and it is denoted as $\mathbf{Z}_{bus}$. The bus impedance matrix is symmetrical.

29. *Name the diagonal elements and off diagonal elements of bus impedance matrix.*

The diagonal elements of bus impedance matrix are called driving point impedances of the buses and off-diagonal elements of bus impedance matrix are called transfer impedances of the buses.

30. *What are the methods available for forming bus impedance matrix?*

The following two methods are available for forming bus impedance matrix.

**Method 1** : Form the bus admittance matrix and then take its inverse to get bus impedance matrix.

**Method 2** : Directly form the bus impedance matrix from the reactance diagram. This method utilizes the techniques of modifications of existing bus impedance matrix due to addition of new bus.

31. *Write the four ways of adding an impedance to an existing system so as to modify bus impedance matrix.*

To modify a bus impedance matrix, a branch of impedance $Z_b$ can be added to original system in the following four different ways.

**Case 1** : Adding a branch of impedance $Z_b$ from a new bus-p to the reference bus.

**Case 2** : Adding a branch of impedance $Z_b$ from a new bus-p to an existing bus-q.

**Case 3** : Adding a branch of impedance $Z_b$ from an existing bus-q to the reference bus.

**Case 4** : Adding a branch of impedance $Z_b$ between two existing buses h and q.

32. *How the $Z_{bus}$ is modified when a branch of impedance $Z_b$ is added from a new bus-p to the reference bus?*

When a branch of impedance $Z_b$ is added from a new bus-p to the reference bus, the order of the bus impedance matrix increases by one.

Let the original bus impedance matrix have an order of n and so the new bus impedance matrix have an order of (n+1). The first n×n submatrix of new bus impedance matrix is the original bus impedance matrix. The elements of $(n+1)^{th}$ column and row are all zeros except the diagonal. The $(n+1)^{th}$ diagonal element is the added branch impedance $Z_b$.

33. *How the $Z_{bus}$ is modified when a branch of impedance $Z_b$ is added from a new bus-p to the existing bus-q?*

When a branch of impedance $Z_b$ is added from a new bus-p to the existing bus-q the order of the bus impedance matrix increases by one.

Let the original bus impedance matrix have an order of n and so the new bus impedance matrix have an order of (n+1). The first n×n submatrix of new bus impedance matrix is the original bus impedance matrix. The elements of (n+1)th column are the elements of $q^{th}$ column and the elements of $(n+1)^{th}$ row are the elements of $q^{th}$ row. The $(n+1)^{th}$ diagonal element is given by sum of $Z_{eq}$ and $Z_b$.

34. *Determine $Z_{bus}$ when* $Y_{bus} = \begin{bmatrix} -j1 & j2 \\ j2 & -j5 \end{bmatrix}$

**Solution**

$$Z_{bus} = Y_{bus}^{-1} = \frac{\text{Adjoint of } Y_{bus}}{\text{Determinant of } Y_{bus}}$$

$$\text{Determinant of } Y_{bus} = \begin{vmatrix} -j1 & j2 \\ j2 & -j5 \end{vmatrix} = (-j1)(-j5) - (j2)^2 = -5 + 4 = -1$$

$$\text{Adjoint of } Y_{bus} = \begin{bmatrix} -j5 & -j2 \\ -j2 & -j1 \end{bmatrix}^T = \begin{bmatrix} -j5 & -j2 \\ -j2 & -j1 \end{bmatrix}$$

$$Z_{bus} = -1 \begin{bmatrix} -j5 & -j2 \\ -j2 & -j1 \end{bmatrix} = \begin{bmatrix} j5 & j2 \\ j2 & j1 \end{bmatrix}$$

35. *What are symmetrical components?*

An unbalanced system of N related vectors can be resolved into N systems of balanced vectors. The N-sets of balanced vectors are called symmetrical components. Each set consist of N-vectors which are equal in length and having equal phase angles between adjacent vectors.

36. *Write the symmetrical components of three phase system.*

In a 3-phase system, the three unbalanced vectors (either current or voltage vectors) can be resolved into three balanced system of vectors. They are

1. Positive sequence components.
2. Negative sequence components
3. Zero sequence components.

37. *What are positive sequence components?*

The positive sequence components of a 3-phase unbalanced vectors consists of three vectors of equal magnitude, displaced from each other by 120° in phase and having the same phase sequence as the original vectors.



38. *What are negative sequence components?*

The negative sequence components of a 3-phase unbalanced vectors consists of three vectors of equal magnitude displaced from each other by 120° in phase and having the phase sequence opposite to that of the original vectors.

39. *What are zero sequence components?*

The zero sequence components of a 3-phase unbalanced vectors consists of 3-vectors of equal magnitude and with zero phase displacement from each other.

40. Express the value of the operator "a" and "$a^2$" in both polar and rectangular form.

$a = 1\angle 120°$ – polar form

$= -0.5 + j0.866$ – Rectangular form

$a^2 = 1\angle 240°$ – polar form

$= -0.5 - j0.866$ – Rectangular form

***Note :*** $a^2$ is conjugate of a.

41. *Prove that $1 + a + a^2 = 0$.*

**Solution**

$$a = 1\angle 120° = -0.5 + j0.866$$

$$a^2 = 1\angle 240° = -0.5 - j0.866$$

$$\therefore 1 + a + a^2 = 1 - 0.5 + j0.866 - 0.5 - j0.866 = 0$$

42. *Express the unbalanced voltages $V_a$, $V_b$ and $V_c$ in terms of symmetrical components $V_{a1}$, $V_{a2}$, and $V_{a0}$.*

**Ans :**

$$V_a = V_{a0} + V_{a1} + V_{a2}$$
$$V_b = V_{a0} + a^2 V_{a1} + a V_{a2}$$
$$V_c = V_{a0} + a V_{a1} + a^2 V_{a2}$$

$$\text{or} \begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix}$$

43. *Express the symmetrical components $V_{a0}$, $V_{a1}$, $V_{a2}$ in terms of unbalanced vectors $V_a$, $V_b$, $V_c$.*

$$V_{a0} = \frac{1}{3}(V_a + V_b + V_c)$$
$$V_{a1} = \frac{1}{3}(V_a + aV_b + a^2 V_c)$$
$$V_{a2} = \frac{1}{3}(V_a + a^2 V_b + aV_c)$$

$$\text{or} \begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix}$$

44. *If $I_a = 12\angle 0°A$, $I_b = 10\angle 90°A$ and $I_c = 10\angle -90°A$. Find the zero sequence current.*

**Solution**

The zero sequence current, $I_{a0} = \frac{1}{3}(I_a + I_b + I_c)$

$$= \frac{1}{3}(12\angle 0° + 10\angle 90° + 10\angle -90°)$$

$$= \frac{1}{3}(12 + j10 - j10) = \frac{12}{3} = 4$$

$$\therefore I_{a0} = 4\angle 0° \text{ A}$$

45. *If $I_a = 18\angle 0°A$, $I_b = 10\angle -30°A$ and $I_c = 10\angle 30°A$. Find the positive sequence current.*

**Solution**

The positive sequence current, $I_{a1} = \frac{1}{3}(I_a + aI_b + a^2 I_c)$

$$= \frac{1}{3}[18\angle 0° + (1\angle 120° \times 10\angle -30°) + (1\angle 240° \times 10\angle 30°)]$$

$$= \frac{1}{3}[18\angle 0° + 10\angle 90° + 10\angle 270°]$$

$$= \frac{1}{3}[18 + j10 - j10] = \frac{18}{3} = 6$$

$$\therefore I_{a1} = 6\angle 0°$$

46. *What are sequence impedance and sequence networks ?*

The sequence impedances are the impedances offered by the devices or components for the like sequence component of the current.

The single phase equivalent circuit of a power system consisting of impedances to current of any one sequence only is called sequence network.

47. *What is meant by positive, negative and zero sequence impedances?*

The impedance of a circuit element for positive, negative and zero sequence component currents are called positive, negative and zero sequence impedances respectively.

48. *What is meant by positive, negative and zero sequence reactance diagram?*

The reactance diagram of a power system, when formed using positive, negative or zero sequence reactances are called positive, negative and zero sequence reactance diagram respectively.



49. *Draw the zero sequence network of a generator when the neutral is grounded and when it is ungrounded?*

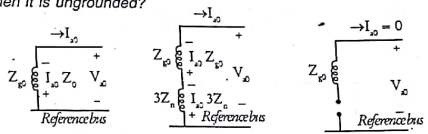



*Fig a : Neutral is solidly grounded* — *Fig b : Neutral is grounded through reactance* — *Fig c : Neutral is ungrounded*

*Fig Q49 : Zero sequence networks of generator*

50. *Draw the zero sequence network of Y/Δ with neutral of star grounded and Δ/Δ transformers.*

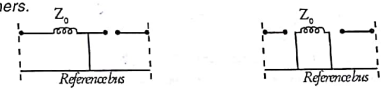



*Fig a : Zero sequence network of Y/Δ transformer with neutral of Y grounded* — *Fig b : Zero sequence network of Δ/Δ transformer*

## 1.10 EXERCISES

### I. State whether the following statements are TRUE/FALSE

1. A balanced 3-phase system is always analysed on per phase basis.
2. The base values for power, voltage, current and impedance can be selected independently.
3. The impedance of a device or component is usually specified in p.u. on the base of name plate rating.
4. In 3-phase system the base kV is phase value and base kVA is per phase kVA.
5. In forming reactance/impedance diagram the base kVA is same for every section but the base kV depends on transformer ratio.
6. The p.u. impedance of a transformer depends on the Y or Δ connection of the winding.
7. In three winding transformer, the three windings may have different kVA rating.
8. In impedance and reactance diagram the neutral impedance has to be included because the return current flows through it.
9. In reactance diagram the induction motors and resistances are neglected.
10. If $Y_{bus}$ is symmetrical then corresponding $Z_{bus}$ is also symmetrical.
11. In matrix partitioning method of bus elimination any bus can be eliminated.
12. The positive sequence components has same phase sequence as that of original vectors.
13. The negative sequence components has same phase sequence as that of original vectors.
14. The induced emfs of synchronous machines are positive sequence voltages.
15. The impedance per phase of transmission line for balanced currents is independent of phase sequence.
16. For all types of transformers the series impedances of all sequences are equal.
17. When neutral is grounded there is path for zero sequence current in transformer.

18. Zero sequence current can flow in one winding of a transformer only when there is a path for it in the other winding.
19. In delta connected winding there is no zero sequence currents.
20. In loads, the zero sequence currents will flow in the network only if a return path exists for it.

**Answers**

| | | | |
|---|---|---|---|
| 1. True | 6. False | 11. False | 16. True |
| 2. False | 7. True | 12. True | 17. False |
| 3. True | 8. False | 13. False | 18. True |
| 4. False | 9. True | 14. True | 19. False |
| 5. True | 10. True | 15. True | 20. True |

## II. Fill in the blanks with appropriate words

1. In a ..... the components of power system are represented by symbols and the interconnections between them are shown by straight lines.
2. For a balanced 3-phase system the ..... is same as positive sequence reactance diagram.
3. The impedance diagrams are used for ..... and reactance diagrams are used for .....
4. The meeting point of various components in a power system is called .....
5. The diagonal point of $Y_{bus}$ are called ..... and off-diagonal elements are called .....
6. The diagonal elements of $Z_{bus}$ are called ..... and off-diagonal elements are called .....
7. An unbalanced system of N related vectors can be resolved into N system of balanced vectors called .....
8. The ..... components consists of three vectors equal in magnitude and phase.
9. In transmission line the ..... reactance is ..... times the positive sequence reactance.
10. The ..... and ..... sequence network of generator will not have any sources.

**Answers**

| | | | |
|---|---|---|---|
| 1. | single line diagram | 6. | driving point impedances, transfer impedances |
| 2. | reactance diagram | 7. | symmetrical components |
| 3. | load flow studies, fault calculations | 8. | zero sequence |
| 4. | bus | 9. | zero sequence, 2 to 3.5 |
| 5. | self admittances, mutual admittances | 10. | negative, zero |

## III. Unsolved problems

1. (a) A generator is rated 30 MVA, 10.5 kV. Its Y-connected winding has a reactance of 1 p.u. Find the ohmic value of the reactance of winding.

   (b) If the generator is working in a circuit for which the bases are specified as 10 MVA, 6.6 kV. Then find the p.u. value of generator working on the specified base.

2. A 15 MVA, 10.5 kV, 3$\phi$ generator has a synchronous reactance of 0.2 p.u. and it is connected to a transmission line through a transformer rated 15 MVA, 33/11 kV with X = 0.15 p.u.

   (i) Calculate the p.u. reactances by taking generator rating as base values.

   (ii) Calculate the p.u. reactances by taking transformer rating as base values.



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3. A single line diagram of an unloaded power system shown in fig E3.1. The generators and transformers are rated as follows. Draw the reactance diagram.

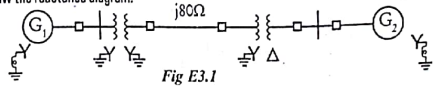



*Fig E3.1*

| | |
|---|---|
| Generator 1 | 50 MVA, 13.8 kV, X" = 0.15 p.u. |
| Generator 2 | 40 MVA, 33 kV, X" = 0.20 p.u. |
| Y-Y transformer | 60 MVA, 16 kV/110 kV, X = 0.1 p.u. |
| Y-Δ transformer | 40 MVA, 33 kV/110 kV, X = 0.15 p.u. |

4. A 40 MVA, 25 kV, 3$\phi$ generator has a subtransient reactance of 25%. It is connected through a Δ-Y transformer to a high voltage transmission line having total series reactance of 50Ω. At the load end of the line is Y-Y step down transformer. Both transformer banks are composed of 1$\phi$ transformer connected for 3$\phi$ operation. Each of the 3 transformer composing each bank is rated 16.67 MVA, 13.8 kV/100 kV with a lekage reactance of 20%. The load represented as impedance, is drawing 30 MVA at 24 kV, 0.9 pf lag. Draw the single line diagram of power network. Choose a base of 30 MVA, 24 kV in the load circuit. Determine also the voltage at the terminals of the generator.

5. Solve the node voltages.

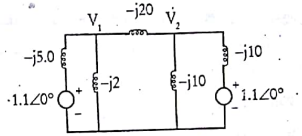



*Fig E5.1*

6. For the network shown in figE6.1. Give the total number of elements, nodes, buses and branches. Write the element of $Y_{bus}$ matrix directly by inspection.

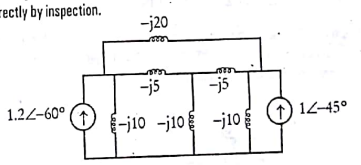



*Fig E6.1*

7. Determine the reduced admittance matrix by eliminating nodes (3) and (4). Values marked in the figE7.1. are p.u. admittances and currents.

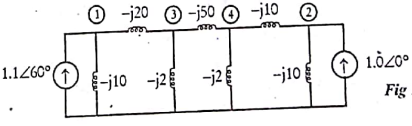



*Fig E7.1*

8. For the system shown in fig E8.1., determine $Z_{bus}$.

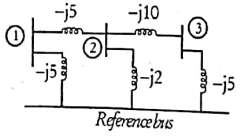



*Fig E8.1*

9. Determine $Z_{bus}$ for the network shown in fig E9.1., where the impedances are given in p.u. Preserve all the 3 nodes.

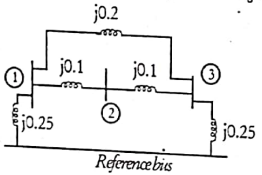



*Fig E9.1*

10. The voltage across a 3$\phi$ unbalanced load are $V_a$ = 200V, $V_b$ = 200 $\angle$180° V and $V_c$ = 600$\angle$145.2 V respectively. Determine the symmetrical components of voltages. Phase sequence is abc.

11. Draw the positive, negative and zero sequence impedance networks for the power system shown in fig E11.1.

Choose a base of 50 MVA, 220 kV in the 50 $\Omega$ transmission line and mark all reactances in p.u. The ratings of the generators and transformers are :

$G_1$ : 25 MVA, 11 kV, X" = 20%

$G_2$ : 25 MVA, 11 kV, X" = 20%

All transformers : 15 MVA, 11/220 kV, X = 10%

The negative sequence reactance of each synchronous machine is equal to its subtransient reactance. The zero sequence reactance of each machine is 8%. Assume that the zero sequence reactances of transmission lines are 250% of their positive sequence reactances.

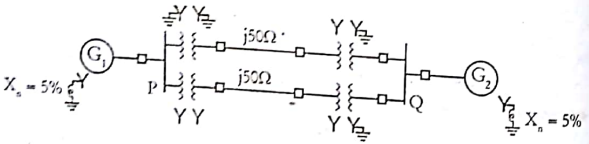
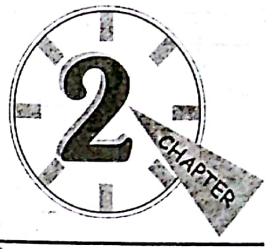



*Fig E11.1*

# LOAD FLOW STUDIES




## 2.1 INTRODUCTION

The load flow studies involves the solution of the power system network under steady state conditions. The solution is obtained by considering certain inequality constraints imposed on node voltages and reactive power of generators.

The main informations obtained from a load-flow study are the magnitude and phase angle of the voltage at each bus and the real and reactive power flowing in each line. The load flow solution also gives the initial conditions of the system when the transient behaviour of the system is to be studied.

The load flow study of a power system is essential to decide the best operation of existing system and for planning the future expansion of the system. It is also essential for designing a new power system.

The load flow solution of power system under balanced load condition have been presented in this chapter. For analysis of balanced system a single phase equivalent circuit of power system such as impedance diagram is adequate.

*A load flow study of a power system generally require the following steps.*

(i) Representation of the system by single line diagram

(ii) Determining the impedance diagram using the informations in single line diagram.

(iii) Formulation of network equations

(iv) Solution of network equations.

Under steady state conditions the network equations will be in the form of simple algebraic equations. In a practical system the load and hence generation are continuously changing, but for calculation purpose we assume that the loads and generation are fixed at a particular value for a suitable period of time, e.g., 15 minutes or 30 minutes.

## III. Unsolved problems

1. For the system shows in fig E3.1, find the fault current in all the branches where a 3-phase fault occurs at bus-3. All reactance are in pu. Neglect prefault current.

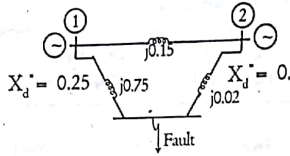



*Fig E 3.1*

2. The system shown in fig E3.2 is initially at no-load. Calculate the subtransient fault current for a 3-phase fault at point F. Take base values as 120 MVA and 69kV.

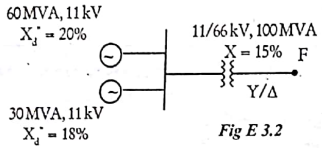



*Fig E 3.2*

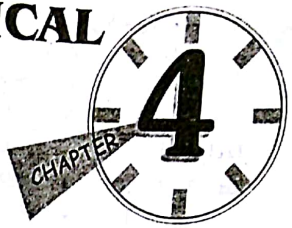
3. A 3-phase transmission line operating at 110kV and having an impedance 5+j20Ω is connected to the generating station through 15,000 kVA step-up transformer. Two alternator are connected to the bus bars. The ratings of the alternator are 10 MVA, 10%, 16kV and 5MVA, 7.5%, 16kV. Calculate the short circuit MVA for a symmetrical fault at the load end of transmission line. Take transformer reactance as 8%.
4. A generator rated at 500 kVA, 440V, with 12% subtransient reactance is supplying a passive load of 400 kW at 0.8 pf lagging. Calculate the initial rms current for a 3ϕ fault at the generator terminals.
5. The bus impedance matrix of a system is given by

$$Z_{bus} = j\begin{bmatrix} 0.14 & 0.06 & 0.08 & 0.09 \\ 0.06 & 0.15 & 0.07 & 0.09 \\ 0.08 & 0.07 & 0.13 & 0.10 \\ 0.09 & 0.09 & 0.10 & 0.15 \end{bmatrix}$$

Generator are connected to buses 1 and 3 and their subtransient reactances were included when finding bus impedance matrix. Find the fault current for a 3-phase fault in bus-3. Also calculate the voltages in all the buses. Find the subtransient fault current in generator at bus-1 if its subtransient reactance is 0.20 pu.

6. Repeat the exercise problem -5 for a 3-phase fault in bus-2.
7. A generator is connected through a circuit breaker to a transformer. The rating of the generator is 40MVA, 12.4kV. Given that $X_d'' = 14\%$, $X_d' = 22\%$ and $X_d = 90\%$. The transformer is rated at 40 MVA, 13/110kV, with a reactance of 10%. For a three phase fault on the HT side of the transformer determine the ratings of circuit breaker.

# UNSYMMETRICAL FAULT ANALYSIS




## 4.1 INTRODUCTION

The unsymmetrical faults are the faults in which the fault currents in the three phases are unequal. The different types of unsymmetrical faults are

i) Single line-to-ground fault
ii) Line-to-line fault
iii) Double line-to-ground fault
iv) One or two open conductor faults..

Since any unsymmetrical fault causes unbalanced currents to flow in the system, the unsymmetrical faults are analyzed using symmetrical components. The concepts of symmetrical components and sequence networks developed in chapter-1 will be used for the analysis of unsymmetrical faults.

### Review of symmetrical components of unbalanced voltages and currents

Let $V_a, V_b, V_c$ be a set of three phase unbalanced voltage vectors.

Let $V_{a1}, V_{b1}, V_{c1}$ = Positive sequence components of voltages
$V_{a2}, V_{b2}, V_{c2}$ = Negative sequence components of voltages
$V_{a0}, V_{b0}, V_{c0}$ = Zero sequence components of voltages

Now the unbalanced voltage vectors are related to sequence component voltages by the following equations.

$$\begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix} \quad \text{.....(4.1)}$$

$$\begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} \quad \text{.....(4.2)}$$

$$V_{a0} = V_{b0} = V_{c0} \quad \text{.....(4.3)}$$

$$V_{b1} = a^2 V_{a1} \quad \& \quad V_{c1} = a V_{a1} \quad \text{.....(4.4)}$$

$$V_{b2} = a V_{a2} \quad \& \quad V_{c2} = a^2 V_{a2} \quad \text{.....(4.5)}$$

$$a = 1\angle 120° \quad \& \quad a^2 = 1\angle 240° \quad \text{.....(4.6)}$$

Let $I_a, I_b, I_c$ be a set of three phase unbalanced current vectors.

Let $I_{a1}, I_{b1}, I_{c1}$ = Positive sequence components of currents

$I_{a2}, I_{b2}, I_{c2}$ = Negative sequence components of currents

$I_{a0}, I_{b0}, I_{c0}$ = Zero sequence components of currents

Now the unbalanced current vectors are related to sequence component currents by the following equations.

$$\begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} \quad \text{.....(4.7)}$$

$$\begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} \quad \text{.....(4.8)}$$

$$I_{a2} = I_{b0} = I_{c0} \quad \text{.....(4.9)}$$

$$I_{b1} = a^2 I_{a1} \quad \& \quad I_{c1} = a I_{a1} \quad \text{.....(4.10)}$$

$$I_{b2} = a I_{a2} \quad \& \quad I_{c2} = a^2 I_{a2} \quad \text{.....(4.11)}$$

$$a = 1\angle 120° \quad \& \quad a^2 = 1\angle 240° \quad \text{.....(4.12)}$$

### Review of sequence networks of a generator

Let $Z_1$ = Positive sequence impedance of generator.

$Z_2$ = Negative sequence impedance of generator.

$Z_{g0}$ = Zero sequence impedance of generator.

$Z_n$ = Zero sequence impedance of generator including neutral reactance.

The positive, negative and zero sequence networks of a generator with neutral grounded through an impedance $Z_n$ are shown in fig 4.1., 4.2. and 4.3 respectively.

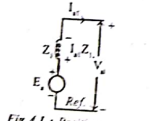



*Fig 4.1 : Positive sequence network of a generator*

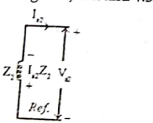



*Fig 4.2 : Negative sequence network of a generator*

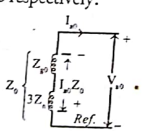



*Fig 4.3 : Zero sequence network of a generator*



From the sequence networks shown above, using kirchoff's laws we can write the following equations

$$V_{a0} = -I_{a0}Z_0 \quad \text{.....(4.13)}$$

$$V_{a1} = E_a - I_{a1}Z_1 \quad \text{.....(4.14)}$$

$$V_{a2} = -I_{a2}Z_2 \quad \text{.....(4.15)}$$

The equations (4.13), (4.14) and (4.15) can be arranged in the form of matrix equation as shown below.

$$\begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix} = \begin{bmatrix} 0 \\ E_a \\ 0 \end{bmatrix} - \begin{bmatrix} Z_0 & 0 & 0 \\ 0 & Z_1 & 0 \\ 0 & 0 & Z_2 \end{bmatrix} \begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} \quad \text{.....(4.16)}$$

The matrix equation(4.16) will be useful in the analysis of unsymmetrical faults on the generator.

## 4.2 SINGLE LINE-TO-GROUND FAULT ON AN UNLOADED GENERATOR

The circuit diagram for a single line-to-ground fault on an unloaded Y-connected generator with its neutral grounded through a reactance is shown in fig 4.4. Here it is assumed that phase-a is shorted to ground. Now the fault current, $I_f = I_a$. Since the generator is unloaded the current in other phases are zero.

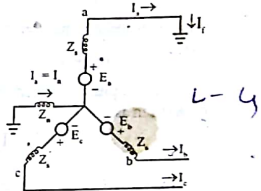



*Fig 4.4 : Circuit diagram for single line-to-ground fault on phase-a of an unloaded generator*

The condition at the fault is expressed by the following equations

$$I_b = 0 \; ; \; I_c = 0 \; ; \; V_a = 0 \quad \text{.....(4.17)}$$

The symmetrical components of the currents are given by

$$\begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} \quad \text{.....(4.18)}$$

On substituting the conditions $I_b = I_c = 0$ in the symmetrical components of currents (equ 4.18) we get,

$$\begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} I_a \\ 0 \\ 0 \end{bmatrix} \quad \text{.....(4.19)}$$

On multiplying the above matrix equation (4.19) we get,

$$I_{a0} = I_{a1} = I_{a2} = \frac{I_a}{3} \quad \text{.....(4.20)}$$

From the sequence networks of the generator we get the following matrix equation [Refer equ(4.16)].

$$\begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix} = \begin{bmatrix} 0 \\ E_a \\ 0 \end{bmatrix} - \begin{bmatrix} Z_0 & 0 & 0 \\ 0 & Z_1 & 0 \\ 0 & 0 & Z_2 \end{bmatrix} \begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} \quad \text{.....(4.21)}$$

On substituting $I_{a0} = I_{a1}$ and $I_{a2} = I_{a1}$ in equ(4.21) we get,

$$\begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix} = \begin{bmatrix} 0 \\ E_a \\ 0 \end{bmatrix} - \begin{bmatrix} Z_0 & 0 & 0 \\ 0 & Z_1 & 0 \\ 0 & 0 & Z_2 \end{bmatrix} \begin{bmatrix} I_{a1} \\ I_{a1} \\ I_{a1} \end{bmatrix} \quad \text{.....(4.22)}$$

On multiplying equ (4.22) we get,

$$V_{a0} = -Z_0 I_{a1} \quad \text{.....(4.23)}$$

$$V_{a1} = E_a - Z_1 I_{a1} \quad \text{.....(4.24)}$$

$$V_{a2} = -Z_2 I_{a1} \quad \text{.....(4.25)}$$

On adding equations (4.23), (4.24) and (4.25) we get,

$$V_{a0} + V_{a1} + V_{a2} = -I_{a1} Z_0 + E_a - I_{a1} Z_1 - I_{a1} Z_2 \quad \text{.....(4.26)}$$

Since phase-a is shorted to ground, $V_a = V_{a0} + V_{a1} + V_{a2} = 0$ and so the equ(4.26) can be written as

$$-I_{a1} Z_0 + E_a - I_{a1} Z_1 - I_{a1} Z_2 = 0$$

$$I_{a1} = \frac{E_a}{Z_1 + Z_2 + Z_0} \quad \text{.....(4.27)}$$

Using equ (4.27) the equivalent circuit of generator during single line-to ground fault is drawn as shown in fig 4.5. Here the positive, negative and zero sequence networks of the generator are connected in series.

If the neutral of the generator is not grounded, the zero sequence network is open-circuited and $Z_0$ is infinite. Under this condition $I_{a1}$ is zero and so $I_{a2}$ & $I_{a0}$ must be zero. Therefore no path exists for the flow of current in the fault unless the generator neutral is grounded.

Here the fault current is given by,

$$I_f = I_a = 3 I_{a1} \quad \text{.....(4.28)}$$

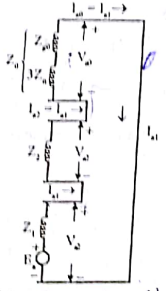



*Fig 4.5 : Connection of sequence network of an unloaded generator for line-to-ground fault on phase -a*



## 4.3 LINE-TO-LINE FAULT ON AN UNLOADED GENERATOR

The circuit diagram for a line-to line fault on an unloaded , Y-connected generator is shown in fig 4.6. Here it is assumed that phase-b and phase c are shorted. Now the fault current $I_b = -I_c$. Since the generator is unloaded the current in phase-a is zero.

The conditions at the fault are expressed by the following equations

$$V_b = V_c \; ; \; I_a = 0 \; ;$$

$$I_b + I_c = 0 \Rightarrow I_b = -I_c \quad \text{.....(4.29)}$$

With $V_c = V_b$, the symmetrical components of voltage are given by

$$\begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} V_a \\ V_b \\ V_b \end{bmatrix} \quad \text{.....(4.30)}$$

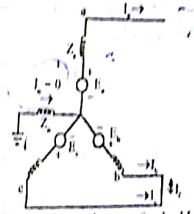



*Fig 4.6 : Circuit diagram for double line to ground fault between phase b & c in an unloaded generator*

On multiplying row 2 & 3 of equ(4.30) we get,

$$V_{a1} = \frac{1}{3}(V_a + aV_b + a^2 V_b) \quad \text{.....(4.31)}$$

$$V_{a2} = \frac{1}{3}(V_a + a^2 V_b + aV_b) \quad \text{.....(4.32)}$$

From equations (4.31) and (4.32) we can write

$$V_{a1} = V_{a2} \quad \text{.....(4.33)}$$

The symmetrical components of current are given by

$$\begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} \quad \text{.....(4.34)}$$

On substituting $I_b = -I_c$ and $I_a = 0$ in equ(4.34) we get,

$$\begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} 0 \\ -I_c \\ I_c \end{bmatrix} \quad \text{.....(4.35)}$$

On multiplying equ(4.35) we get,

$$I_{a0} = \frac{1}{3}[-I_c + I_c] = 0 \quad \text{.....(4.36)}$$

$$I_{a1} = \frac{1}{3}[-a I_c + a^2 I_c]$$

$$I_{a2} = \frac{1}{3}\left[-a^2 I_c + a I_c\right] = -\frac{1}{3}\left[-a I_c + a^2 I_c\right] \quad \text{.....(4.37)}$$

From equations (4.36) and (4.37) we can write

$$I_{a2} = -I_{a1} \quad \text{.....(4.38)}$$

From the sequence networks of the generator we get the following matrix equations. [Refer equ(4.16)].

$$\begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix} = \begin{bmatrix} 0 \\ E_a \\ 0 \end{bmatrix} - \begin{bmatrix} Z_0 & 0 & 0 \\ 0 & Z_1 & 0 \\ 0 & 0 & Z_2 \end{bmatrix} \begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} \quad \text{.....(4.39)}$$

On substituting $I_{a0} = 0$ and $I_{a2} = -I_{a1}$ in equ (4.39) we get,

$$\begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix} = \begin{bmatrix} 0 \\ E_a \\ 0 \end{bmatrix} - \begin{bmatrix} Z_0 & 0 & 0 \\ 0 & Z_1 & 0 \\ 0 & 0 & Z_2 \end{bmatrix} \begin{bmatrix} 0 \\ I_{a1} \\ -I_{a1} \end{bmatrix} \quad \text{.....(4.40)}$$

On multiplying equ (4.40) we get,

$$V_{a0} = 0$$

$$V_{a1} = E_a - Z_1 I_{a1}$$

$$V_{a2} = Z_2 I_{a1} \quad \text{.....(4.41)}$$

From equ(4.33) we know that $V_{a1} = V_{a2}$. Hence, on equating (4.41) and (4.42) we get, .....(4.42)

$$E_a - Z_1 I_{a1} = Z_2 I_{a1}$$

(or) $I_{a1}(Z_1 + Z_2) = E_a$

$$\therefore I_{a1} = \frac{E_a}{Z_1 + Z_2} \quad \text{.....(4.43)}$$

Using equ (4.43), the equivalent circuit of generator during single line to ground fault is drawn as shown in fig (4.7).

Since $V_{a1} = V_{a2}$, here the positive and negative sequence networks of the generator must be in parallel. Since $V_{a0} = 0$, the zero sequence network is shorted and so it need not be considered. (or Since $Z_b$ does not enter into the equations, the zero-sequence network is not used). Since this type of fault does not involve ground, the neutral current $I_n = 0$. Hence the presence or absence of a grounded neutral at the generator does not affect the fault current..

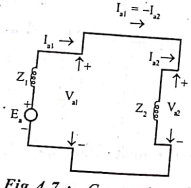



*Fig 4.7 : Connection of the sequence networks of an unloaded generator for a line-to-line fault between phases b and c*



Here the fault current, $I_f = I_b = -I_c$

The unbalanced currents $I_a$, $I_b$ and $I_c$ are related to the symmetrical components of currents by the following matrix equation.

$$\begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} \quad \text{.....(4.44)}$$

From equation (4.44) we get,

$$I_b = I_{a0} + a^2 I_{a1} + a I_{a2} \quad \text{.....(4.45)}$$

On substituting the conditions $I_{a0} = 0$ and $I_{a2} = -I_{a1}$ which exists during line-to-line fault in equ(4.45) we get,

$$I_b = a^2 I_{a1} - a I_{a1} = I_{a1}(a^2 - a)$$

$\therefore$ Fault current, $I_f = I_b = I_{a1}(a^2 - a)$ .....(4.46)

## 4.4 DOUBLE LINE-TO -GROUND FAULT ON AN UNLOADED GENERATOR

The circuit diagram for a double line-to-ground fault on an unloaded Y connected generator having a grounded neutral is shown in fig 4.8. The faulted phases are b and c. Now the fault current, $I_f = I_b + I_c$. Since the generator is unloaded the current in phase-a is zero.

The conditions at the fault are expressed by the following equations

$$V_b = 0, \quad V_c = 0, \quad I_a = 0.$$

With $V_b = V_c = 0$, the symmetrical components of voltage are given by

$$\begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} V_a \\ 0 \\ 0 \end{bmatrix} \quad \text{.....(4.47)}$$

On multiplying equ(4.47) we get,

$$V_{a0} = V_{a1} = V_{a2} = \frac{V_a}{3} \quad \text{.....(4.48)}$$

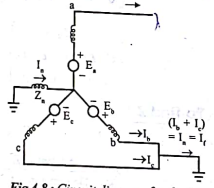



*Fig 4.8 : Circuit diagram for double line-to-ground fault between phase b & c in an unloaded generator*

e networks of the generator we get the following matrix equation.

$$= \begin{bmatrix} Z_0 & 0 & 0 \\ 0 & Z_1 & 0 \\ 0 & 0 & Z_2 \end{bmatrix} \begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} \quad \text{.....(4.49)}$$

row-2 of equ(4.49) we get,

$$= Z_1 \quad \text{.....(4.50)}$$

get $V_{a0} - V_{a1} - V_{a2}$ and from equ(4.50) we get $V_{a1} = E_a - I_{a1} Z_1$. On s in equ(4.49) we get,

$$\begin{bmatrix} 0 \\ E_a \\ 0 \end{bmatrix} - \begin{bmatrix} Z_0 & 0 & 0 \\ 0 & Z_1 & 0 \\ 0 & 0 & Z_2 \end{bmatrix} \begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} \quad \text{.....(4.51)}$$

u(4.51) we get,

$$\begin{bmatrix} \\ \\ \end{bmatrix} = \begin{bmatrix} 0 \\ E_a \\ 0 \end{bmatrix} - \begin{bmatrix} E_a - I_{a1} Z_1 \\ E_a - I_{a1} Z_1 \\ E_a - I_{a1} Z_1 \end{bmatrix} \quad \text{.....(4.52)}$$

$$\begin{bmatrix} 0 \\ 0 \\ Z_2 \end{bmatrix} \quad \therefore Z^{-1} = \begin{bmatrix} 1/Z_0 & 0 & 0 \\ 0 & 1/Z_1 & 0 \\ 0 & 0 & 1/Z_2 \end{bmatrix}$$

$$\frac{Z}{\text{f } Z} = \frac{Z^T_{cof.}}{\Delta Z}$$

$Z_0 Z_1 Z_2$

$$\begin{bmatrix} 0 \\ 0 \\ Z_0 Z_1 \end{bmatrix}^T = \begin{bmatrix} Z_1 Z_2 & 0 & 0 \\ 0 & Z_0 Z_2 & 0 \\ 0 & 0 & Z_0 Z_1 \end{bmatrix}$$

$$\begin{bmatrix} Z_2 & 0 & 0 \\ & Z_0 Z_2 & 0 \\ & 0 & Z_0 Z_1 \end{bmatrix} = \begin{bmatrix} 1/Z_0 & 0 & 0 \\ 0 & 1/Z_1 & 0 \\ 0 & 0 & 1/Z_2 \end{bmatrix}$$



On premultiplying the equ (4.52) by $Z^{-1}$ we get,

$$\begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} = \begin{bmatrix} 1/Z_0 & 0 & 0 \\ 0 & 1/Z_1 & 0 \\ 0 & 0 & 1/Z_2 \end{bmatrix} \begin{bmatrix} 0 \\ E_a \\ 0 \end{bmatrix} - \begin{bmatrix} 1/Z_0 & 0 \\ 0 & 1/Z \\ 0 & 0 \end{bmatrix}$$

$$\therefore \begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} = \begin{bmatrix} 0 \\ \frac{E_a}{Z_1} \\ 0 \end{bmatrix} - \begin{bmatrix} \frac{E_a - I_{a1} Z_1}{Z_0} \\ \frac{E_a - I_{a1} Z_1}{Z_1} \\ \frac{E_a - I_{a1} Z_1}{Z_2} \end{bmatrix}$$

On multiplying equ(4.53) we get

$$I_{a0} = -\left(\frac{E_a - I_{a1} Z_1}{Z_0}\right) = -\frac{E_a}{Z_0} + \frac{I_{a1} Z_1}{Z_0}$$

$$I_{a1} = \frac{E_a}{Z_1} - \left(\frac{E_a - I_{a1} Z_1}{Z_1}\right) = \frac{E_a}{Z_1} - \frac{E_a}{Z_1} + \frac{I_{a1} Z_1}{Z_1} = I_{a1}$$

$$I_{a2} = -\left(\frac{E_a - I_{a1} Z_1}{Z_2}\right) = -\frac{E_a}{Z_2} + \frac{I_{a1} Z_1}{Z_2}$$

Here $I_a = 0$, On expressing $I_a$ by its symmetrical compo

$$I_a = I_{a0} + I_{a1} + I_{a2} = 0$$

On substituting for $I_{a0}$, $I_{a1}$ and $I_{a2}$ from equations (4.54), (4.5

$$-\frac{E_a}{Z_0} + \frac{I_{a1} Z_1}{Z_0} + I_{a1} - \frac{E_a}{Z_2} + \frac{I_{a1} Z_1}{Z_2} = 0$$

$$I_{a1}\left(\frac{Z_1}{Z_0} + 1 + \frac{Z_1}{Z_2}\right) = \frac{E_a}{Z_0} + \frac{E_a}{Z_2}$$

$$I_{a1}\left(1 + Z_1\left(\frac{1}{Z_0} + \frac{1}{Z_2}\right)\right) = E_a\left(\frac{1}{Z_0} + \frac{1}{Z_2}\right)$$

$$I_{a1}\left(1 + Z_1\left(\frac{Z_0 + Z_2}{Z_0 Z_2}\right)\right) = E_a\left(\frac{Z_0 + Z_2}{Z_0 Z_2}\right)$$

On multiplying the equ(4.58) throughout by $\frac{Z_0 Z_2}{Z_0 + Z_2}$ we g

$$I_{a1}\left(\frac{Z_0 Z_2}{Z_0 + Z_2} + Z_1\right) = E_a$$

$$\therefore I_{a1} = \frac{E_a}{Z_1 + \frac{Z_0 Z_2}{Z_0 + Z_2}} \quad \text{.....(4.59)}$$

Using equ(4.59) the equivalent circuit of generator under double line-to-ground fault can be drawn as shown in fig 4.9.

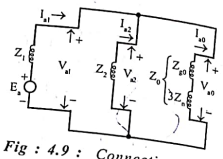


*Fig : 4.9 : Connection of the sequence networks of an unloaded generator for a double line-to-ground fault on phase b and c.*

Under this fault conditon the sequence networks should be connected in parallel since the positive, negative, and zero-sequence voltages are equal during this fault. In the absence of a ground connection at the generator no current can flow into the ground at the fault. In this case $Z_0$ would be infinite and $I_{a0}$ would be zero and so the fault will be similar to line-to-line fault.

Here the fault current, $I_f = I_b + I_c$ ....(4.60)

To determine the fault current, first the symmetrical components $I_{a1}$, $I_{a0}$ and $I_{a2}$ are calculated using equations (4.59), (4.54) and (4.56). These equations are presented here for convenience

$$I_{a1} = \frac{E_a}{Z_1 + \frac{Z_0 Z_2}{Z_0 + Z_2}} \quad ; \quad I_{a0} = -\frac{E_a}{Z_0} + \frac{I_{a1} Z_1}{Z_0} \quad ; \quad I_{a2} = -\frac{E_a}{Z_2} + \frac{I_{a1} Z_1}{Z_2}$$

Alternatively the $I_{a1}$ can be calculated using equ(4.59) and then $I_{a2}$ & $I_{a0}$ can be calculated using current division rule as shown below

$$I_{a2} = -I_{a1} \times \frac{Z_0}{Z_0 + Z_2} \quad \& \quad I_{a0} = -(I_{a1} - I_{a2})$$

The unbalanced currents $I_a$, $I_b$ and $I_c$ are related to the symmetrical components of currents by the following equation.

$$\begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} \quad \text{.....(4.61)}$$

From equation (4.61) we get,

$$I_b = I_{a0} + a^2 I_{a1} + a I_{a2}$$

$$I_c = I_{a0} + a I_{a1} + a^2 I_{a2}$$

$$\therefore \text{Fault current, } I_f = I_b + I_c = I_{a0} + a^2 I_{a1} + a I_{a2} + I_{a0} + a I_{a1} + a^2 I_{a2}$$

$$= 2I_{a0} + (a + a^2) I_{a1} + (a + a^2) I_{a2}$$

$$= 2I_{a0} + (a + a^2)(I_{a1} + I_{a2}) \quad \text{.....(4.62)}$$



## 4.5 UNSYMMETRICAL FAULTS ON POWER SYSTEMS

The unsymmetrical faults on the power system are analyzed using thevenin's equivalent of positive, negative and zero sequence networks of the power system. The thevenin's equivalent is obtained with respect to fault point and reference.

In power system the prefault voltages and currents are assumed to be balanced. Hence the prefault voltage at the fault point is a balanced voltage and so it has only positive sequence component. The negative and zero sequence component of prefault voltage at the fault point is zero. Therefore the prefault voltage at the fault point is the thevenin's voltage of positive sequence network.

Let $Z_1$ = Thevenin's impedance of positive sequence network.
$Z_2$ = Thevenin's impedance of negative sequence network.
$Z_0$ = Thevenin's impedance of zero sequence network.
$V_{pf}$ = Prefault voltage at the fault point.

Thevenin's equivalent of positive, negative and zero sequence networks of the power system with respect to fault point will be as shown in fig (4.10), (4.11) and (4.12) respectively.

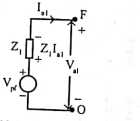


*Fig 4.10 : Thevenin's equivalent of positive sequence network*

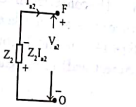


*Fig 4.11 : Thevenin's equivalent of negative sequence network*

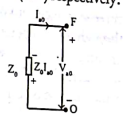


*Fig 4.12 : Thevenin's equivalent of zero sequence network*

From fig (4.10), (4.11) and (4.12) using kirchoff's voltage law we get,

$$V_{a0} = -Z_0 I_{a0} \quad \text{.....(4.63)}$$

$$V_{a1} = V_{pf} - Z_1 I_{a1} \quad \text{.....(4.64)}$$

$$V_{a2} = -Z_2 I_{a2} \quad \text{.....(4.65)}$$

On arranging the equations in the matrix form we get,

$$\begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix} = \begin{bmatrix} 0 \\ V_{pf} \\ 0 \end{bmatrix} - \begin{bmatrix} Z_0 & 0 & 0 \\ 0 & Z_1 & 0 \\ 0 & 0 & Z_2 \end{bmatrix} \begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} \quad \text{.....(4.66)}$$

The matrix equation (4.66) will be useful in the analysis of unsymmetrical faults on the power system.

## Single line - to Ground Fault

A single line-to-ground fault at point F in a power system through a fault impedance $Z_f$ can be represented by connecting three stubs as shown in fig 4.19. The fault is on phase a.

At the fault point F, the following relations exist

$I_b = 0$; $I_c = 0$ ; $V_a = Z_f I_a$

The symmetrical components of the fault current are

$$\begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix}\begin{bmatrix} I_a \\ 0 \\ 0 \end{bmatrix} \quad \text{.....(4.76)}$$

$$\therefore I_{a1} = I_{a2} = I_{a0} = \frac{1}{3} I_a \quad \text{.....(4.77)}$$

(or) $I_a = 3 I_{a1}$ .....(4.78)

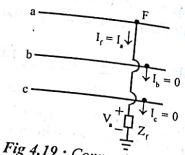



*Fig 4.19 : Connection diagram of stubs for a single line-to ground fault through an impedance*

Expressing the voltage $V_a$ in terms of symmetrical components, we have

$V_a = V_{a1} + V_{a2} + V_{a0}$ .....(4.79)

Using equations (4.76) ,(4.78) and (4.79) we can write

$V_{a1} + V_{a2} + V_{a0} = Z_f I_a = 3Z_f\, I_{a1}$ .....(4.80)

From equations (4.77) and (4.80) we can say that all sequence currents are equal and the sum of sequence voltages equals $3Z_f\, I_{a1}$. Therefore, these equations suggest a series connection of sequence networks through an impedance $3Z_f$ as shown in fig 4.20.

From fig 4.20 we can write

$$I_{a1} = \frac{V_{pf}}{(Z_1 + Z_2 + Z_0) + 3Z_f} \quad \text{.....(4.81)}$$

Fault current, $I_f = I_a = 3I_{a1} = \dfrac{3V_{pf}}{(Z_1 + Z_2 + Z_0) + 3Z_f}$ .....(4.82)

The symmetrical components of phase-a voltage are calculated by writing kirchoff's voltage law equations in the sequence networks of fig 4.20 as shown below.

$V_{a1} = V_{pf} - Z_1 I_{a1}$

$V_{a2} = -Z_2\, I_{a1}$

$V_{a0} = -Z_0 I_{a1}$

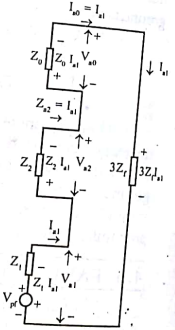



*Fig 4.20 Connection of sequence networks for a single line to ground fault through an impedance $Z_f$*



From the symmetrical components of phase-a voltage, the phase voltages $V_a$, $V_b$ and $V_c$ are calculated using the following matrix equation.

$$\begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix}\begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix}$$

## Line - to - Line Fault

A line-to-line fault at point F in a power system between phases b and c through a fault impedance $Z_f$ can be represented by connecting three stubs as shown in fig 4.21. The currents and voltages at the fault can be expressed as

$I_a = 0$ and $I_b = -I_c$ .....(4.83)

Also $V_b - V_c = I_b Z_f$ ; $\therefore V_c = V_b - I_b Z_f$ .....(4.84)

The symmetrical components of the currents after substituting $I_a = 0$ and $I_c = -I_b$ we get,

$$\begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix}\begin{bmatrix} 0 \\ I_b \\ -I_b \end{bmatrix} \quad \text{.....(4.85)}$$

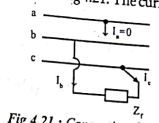



*Fig 4.21 : Connection diagram of stubs for a line-to-line fault through an impedance*

On multiplying equ(4.85) we get,

$$I_{a0} = \frac{1}{3}(I_b - I_b) = 0$$

$$I_{a1} = \frac{1}{3}(aI_b - a^2 I_b)$$

$$I_{a2} = \frac{1}{3}(a^2 I_b - aI_b) = -I_{a1} \quad \text{.....(4.86)}$$

The line currents can be obtained from the following matrix equation, after substituting $I_{a0} = 0$ and $I_{a2} = -I_{a1}$

$$\begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix}\begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix}\begin{bmatrix} 0 \\ I_{a1} \\ -I_{a1} \end{bmatrix} \quad \text{.....(4.87)}$$

From row -2 of matrix equ(4.87) we get,

$$I_b = a^2 I_{a1} - aI_{a1} = I_{a1}(a^2 - a) = I_{a1}\left(-\frac{1}{2} - j\frac{\sqrt{3}}{2} + \frac{1}{2} - j\frac{\sqrt{3}}{2}\right) = -j\sqrt{3}\, I_{a1} \quad \text{.....(4.88)}$$

The symmetrical components of phase voltages are

$$\begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix}\begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} \quad \text{—(4.89)}$$

On substituting $V_c = V_b - I_b Z_f$ in equ (4.89) we get,

$$\begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix}\begin{bmatrix} V_a \\ V_b \\ V_b - I_b Z_f \end{bmatrix} \quad \text{.....(4.90)}$$

From row-2 of matrix equ (4.90) we get,

$$V_{a1} = \frac{1}{3}[V_a + aV_b + a^2V_b - a^2 I_b Z_f]$$

$$\therefore 3V_{a1} = V_a - V_b(a - a^2) - a^2 Z_f I_b \quad \text{—(4.91)}$$

From row-3 of matrix equ (4.90) we get,

$$V_{a2} = \frac{1}{3}[V_a + a^2V_b + aV_b - aI_b Z_f]$$

$$\therefore 3V_{a2} = V_a - V_b(a^2 - a) - a Z_f I_b \quad \text{—(4.92)}$$

On subtracting equ(4.92) from equ(4.91) we get,

$$3V_{a1} - 3V_{a2} = V_a - V_b(a - a^2) - a^2 Z_f I_b - V_a - V_b(a^2 - a) - aZ_f I_b$$

$$= (-a^2 - a) Z_f I_b$$

$$= \left(\frac{1}{2} - j\frac{\sqrt{3}}{2} - \frac{1}{2} - j\frac{\sqrt{3}}{2}\right) Z_f I_b = -j\sqrt{3} Z_f I_b$$

$$\therefore V_{a1} - V_{a2} = \frac{1}{3} j\sqrt{3} Z_f I_b = j\frac{1}{\sqrt{3}} Z_f I_b \quad \text{.....(4.93)}$$

On substituting for $I_b$ from equ (4.88) in equ(4.93) we get,

$$V_{a1} - V_{a2} = j\frac{1}{\sqrt{3}} Z_f(-j\sqrt{3} I_{a1}) = Z_f I_{a1}$$

$$\therefore V_{a1} - V_{a2} = Z_f I_{a1} \quad \text{.....(4.94)}$$

The equations (4.89) and (4.94) suggest parallel connection of positive and negative sequence networks through a series impedance $Z_f$. Therefore the equivalent circuit of power system under-line-to line fault can be drawn as shown in fig 4.22.



From the fig 4.22 we get

$$I_{a1} = \frac{V_{pf}}{Z_1 + Z_2 + Z_f} \quad \text{.....(4.95)}$$

The fault current, $I_f = I_b = -j\sqrt{3}\, I_{a1}$ .....(4.96)
(Refer equation 4.88)

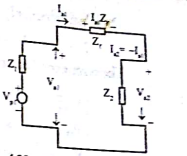



*Fig 4.22: Connection of sequence networks for a line-to-line fault through an impedance $Z_f$.*

## Double Line - to - Ground Fault

A double line-to-ground fault at point F in a power system, through a fault impedance $Z_f$ can be represented by connecting three stubs as shown in fig 4.23.

The current and voltage conditions at the fault are

$$I_a = 0 \quad \text{.....(4.97)}$$

$$V_b = V_c = Z_f(I_b + I_c) \quad \text{.....(4.98)}$$

The line currents are given by

$$\begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix}\begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} \quad \text{.....(4.99)}$$

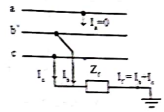



*Fig 4.23 :Connection diagram of stubs for a double line-to-ground fault through an impedance*

From row-1 of equ (4.99) we get,

$$I_a = I_{a0} + I_{a1} + I_{a2}$$

From equ (4.97) we get $I_a = 0$

$$\therefore I_{a0} + I_{a1} + I_{a2} = 0 \quad \text{(or)} \quad I_{a0} = -(I_{a1} + I_{a2}) \quad \text{—(4.100)}$$

From row-2 and 3 of equ (4.99) we get,

$$I_b = I_{a0} + a^2 I_{a1} + a I_{a2}$$

$$I_c = I_{a0} + a I_{a1} + a^2 I_{a2}$$

$$I_b + I_c = I_{a0} + a^2 I_{a1} + a I_{a2} + I_{a0} + a I_{a1} + a^2 I_{a2}$$

$$\therefore \quad = 2 I_{a0} + (a^2 + a) I_{a1} + (a^2 + a) I_{a2} \quad \text{—(4.101)}$$

We know that, $1 + a + a^2 = 0$, $\quad \therefore a + a^2 = -1 \quad \text{—(4.102)}$

From equ(4.101) and (4.102) we get,

$$I_b + I_c = 2I_{a0} - I_{a1} - I_{a2} \quad \text{—(4.103)}$$

$$= 2I_{a0} - (I_{a1} + I_{a2})$$

On substituting for $I_{a1} + I_{a2}$ from equ (4.100) in equ (4.103) we get,

$$I_b + I_c = 2I_{a0} - (-I_{a0}) = 3I_{a0} \quad \text{.....(4.104)}$$

The symmetrical components of voltages after substituting $V_c = V_b$ are given by,

$$\begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix}\begin{bmatrix} V_a \\ V_b \\ V_b \end{bmatrix} \quad \text{.....(4.105)}$$

From row-1 of equ 4.105 we get,

$$V_{a0} = \frac{1}{3}[V_a + V_b + V_b] = \frac{1}{3}(V_a + 2V_b) \quad \text{..... (4.106)}$$

From row-2 and 3 of equ(4.105) we get,

$$V_{a1} = V_{a2} = \frac{1}{3}[V_a + aV_b + a^2V_b] \qquad (\because a + a^2 = -1)$$

$$= \frac{1}{3}(V_a + (a + a^2)V_b) = \frac{1}{3}(V_a - V_b) \quad \text{....(4.107)}$$

From equation (4.106) and (4.107) we can write,

$$V_{a0} - V_{a1} = \frac{1}{3}(V_a + 2V_b) - \frac{1}{3}(V_a - V_b)$$

$$= \frac{1}{3}(V_a + 2V_b - V_a + V_b) = V_b$$

$$\therefore V_{a0} - V_{a1} = V_b$$

On substituting for $V_b$ from equ(4.98) in equ(4.108) we get, .....(4.108)

$$V_{a2} - V_{a1} = Z_f(I_b - I_c)$$

On substituting $(I_b + I_c)$ from equ (4.104) in equ(4.109) we get, .....(4.109)

$$V_{a2} - V_{a1} = Z_f 3I_{a2}$$

$$\therefore V_{a2} = V_{a1} + 3Z_f I_{a2} \quad \text{.....(4.100)} \qquad (\because V_{a2} = V_{a2})$$

Also, $V_{a2} = V_{a2} + 3Z_f I_{a0}$ .....(4.101)

Equations (4.100) and (4.110) suggest the connection of sequence networks as shown in fig 4.24 for a double line-to-ground fault.

From fig 4.24 we can write,

$$I_{a1} = \frac{V_{pf}}{Z_1 + \dfrac{Z_2(Z_0 + 3Z_f)}{Z_2 + Z_0 + 3Z_f}} \quad \text{.....(4.112)}$$

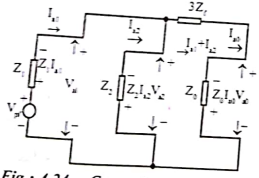



*Fig : 4.24 : Connection of sequence networks for a double line-to-ground fault through an impedance*



$$V_{a1} = V_{pf} - Z_1 I_{a1}$$

$$V_{a2} = V_{a1} \quad ; \quad I_{a2} = \frac{V_{a2}}{Z_2} \quad ; \quad I_{a0} = -(I_{a1} + I_{a2}) \quad \text{.....(4.113)}$$

From fig (4.23) and equ(4.104) we get, .....(4.114)

Fault current, $I_f = I_b + I_c = 3I_{a0}$ .....(4.115)

## EXAMPLE 4.1

Two 11 kV, 20 MVA, three-phase, star connected generators operate in parallel as shown in figure 4.1.1.

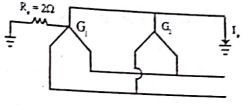



*Fig 4.1.1.*

The positive, negative and zero sequence reactances of each being respectively, j0.18, j0.15, j0.10 p.u. The star point of one of the generator is isolated and that of the other is earthed through a 2.0 Ω resistor. A single line-to-ground fault occurs at the terminals of one of the generators. Estimate (i) fault current, (ii) current in grounded resistor and (iii) voltage across grounding resistor.

### SOLUTION

Let us choose the generator ratings as base values.

$$\therefore MVA_b = 20 MVA \quad ; \quad kV_b = 11 kV$$

$$\text{Base impedance, } Z_b = \frac{kV_b^2}{MVA_b} = \frac{11^2}{20} = 6.05\ \Omega$$

$$\text{p.u. value of neutral resistance} = \frac{\text{Actual value}}{\text{Base impedance}} = \frac{2}{6.05} = 0.3306 \text{ p.u.}$$

The single line diagram and the sequence networks of the given power system are shown in fig (4.1.2) to (4.1.5)

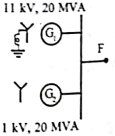



*Fig 4.1.2 : Single line diagram*

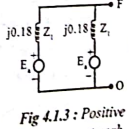



*Fig 4.1.3 : Positive sequence network*

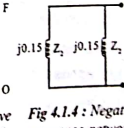



*Fig 4.1.4 : Negative sequence network*

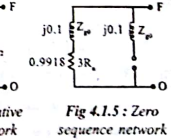



*Fig 4.1.5 : Zero sequence network*

The thevenin's equivalent of the sequence networks are shown in fig (4.1.6) to (4.1.8)

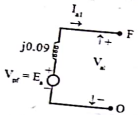



*Fig 4.1.6 : Thevenin's equivalent of p.s. network*

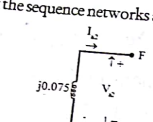



*Fig 4.1.7 : Thevenin's equivalent of n.s. network*

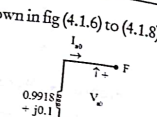



*Fig 4.1.8 : Thevenin's equivalent of z.s. network*

For a single line-to-ground fault,

$$I_{a1} = I_{a2} = I_{a0} \quad \text{and} \quad I_f = I_a = 3\,I_{a1}$$

Hence the thevenin's equivalent of sequence networks are connected in series as shown in fig (4.1.9). The fault current is calculated by taking the prefault voltage $V_{pf} = 1$ p.u.

From fig 4.1.9. we get,

$$I_{a1} = \frac{1\angle 0°}{j0.09 + j0.075 + 0.9918 + j0.1}$$

$$= \frac{1}{0.9918 + j0.265} = \frac{1}{1.0266\angle 15°} = 0.9741\angle -15° \text{ p.u.}$$

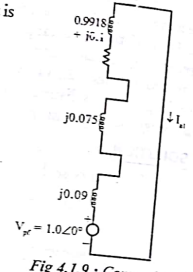



*Fig 4.1.9 : Connection of sequence networks*

### i. To find fault current

Fault current, $I_f = I_a = 3\,I_{a1}$

$= 3 \times 0.9741\angle -15°$

$= 2.9223\angle -15°$ p.u.

Base current, $I_b = \dfrac{kVA_b}{\sqrt{3}\,kV_b} = \dfrac{20 \times 1000}{\sqrt{3} \times 11} = 1049.7$ A

∴ Actual value of fault current = p.u value of fault current × Base current

$= 2.9223\angle -15° \times 1049.7 = 3067.5\angle -15°$ A $= 3.0675\angle -15°$ kA

### ii. To find the current through neutral resistor

The current through the neutral resistor is same as that of fault current.

∴ Current through neutral resistor = $2.9223\angle -15°$ p.u. or $3.0675\angle -15°$ kA.

### iii. To find the voltage across grounding resistor

From the thevenin's equivalent of zero sequence network we get,

The voltage across grounding resistor = $3\,R_n\,I_{a0} = 3\,R_n\,I_{a1}$

$= 3 \times 0.3306 \times 0.9741\angle -15° = 0.9661\angle -15°$ p.u.



Actual value of voltage across grounding resistor = p.u. value of voltage $\times \dfrac{kV_b}{\sqrt{3}}$ $\left(\because kV_b \text{ is line value}\right)$

$= 0.9661\angle -15° \times \dfrac{11}{\sqrt{3}} = 6.1356\angle -15°$ kV

### RESULT

i. Fault current = $2.9223\angle -15°$ p.u. or $3.0675\angle -15°$ kA.

ii. Neutral current = $2.9223\angle -15°$ p.u. or $3.0675\angle -15°$ kA.

iii. Voltage across neutral resistance = $0.9661\angle -15°$ p.u. or $6.1356\angle -15°$ kV.

### EXAMPLE 4.2

A salient - pole generator without dampers is rated 20 MVA, 13.8 kV and has a direct axis subtransient reactance of 0.25 per unit. The negative and zero sequence reactances are 0.35 and 0.10 per unit respectively. The neutral of the generator is solidly grounded. Determine the sub-transient current in the generator and the line-to-line voltages for subtransient conditions when a single line-to-ground fault occurs at the generator terminals with generator operating unloaded at rated voltage. Neglect resistance.

### SOLUTION

Let us choose the generator ratings as base values.

$\therefore MVA_b = 20$MVA ; $kV_b = 13.8$ kV

Base current, $I_b = \dfrac{kVA_b}{\sqrt{3}\,kV_b} = \dfrac{20 \times 1000}{\sqrt{3} \times 13.8} = 836.7$ A

Given that, $Z_1 = j0.25$ p.u. ; $Z_2 = j0.35$ p.u. and $Z_0 = j0.1$ p.u.

For a single line-to-ground fault in an unloaded generator the sequence networks are connected in series as shown in fig 4.2.1. The following relations exists for a single line-to-ground fault.

$$I_{a0} = I_{a1} = I_{a2} \quad ; \quad I_f = I_a = 3I_{a1}$$

$$I_{a1} = \frac{E_a}{Z_1 + Z_2 + Z_0}$$

Here $E_a$ is phase value of induced emf. Let it be 1 p.u. (i.e. $E_a = 1\angle 0°$ p.u.)

$$\therefore I_{a1} = \frac{1}{j0.25 + j0.35 + j0.1} = \frac{1}{j0.7} = -j1.43 = 1.43\angle -90° \text{ p.u.}$$

Subtransient fault current in p.u., $I_f'' = |I_a| = 3\,|I_{a1}| = 3 \times 1.43$

$= 4.29$ p.u.

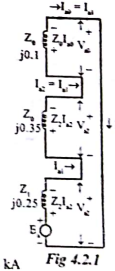



*Fig 4.2.1*

Actual value of subtransient fault current = p.u.value of fault × Base current

$= 4.29 \times 836.7 = 3589.4$ A $= 3.5894$ kA

From the sequence networks of the generator the symmetrical components of phase -a voltage are calculated as shown below.

$$V_{a1} = E_a - I_{a1}Z_1 = 1.0 - (-j1.43)\,(j0.25) = 1.0 - 0.357 = 0.643 \text{ p.u.}$$

$$V_{a2} = -I_{a2}Z_2 = -(-j1.43)\,(j0.35) = -0.50 \text{ p.u.}$$

$$V_{a0} = -I_{a0}Z_0 = -(-j1.43)\,(j0.1) = -0.143 \text{ p.u.}$$

The phase voltages are given by,

$$\begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix}$$

where $a = 1\angle 120° = -0.5 + j0.866$
and $a^2 = 1\angle 240° = -0.5 - j0.866$

$$V_a = V_{a0} + V_{a1} + V_{a2} = 0.643 - 0.50 - 0.143 = 0$$

$$V_b = V_{a0} + a^2 V_{a1} + aV_{a2}$$

$$= -0.143 + 0.643\,(-0.5 - j0.866) - 0.50(-0.5 + j0.866)$$

$$= -0.143 - 0.322 - j0.557 + 0.25 - j0.433 = -0.215 - j0.990 \text{ p.u.}$$

$$V_c = V_{a0} + aV_{a1} + a^2 V_{a2}$$

$$= -0.143 + 0.643(-0.5 + j0.866) - 0.5(-0.5 - j0.866)$$

$$= -0.143 - 0.322 + j0.557 + 0.25 + j0.433 = -0.215 + j0.99 \text{ p.u.}$$

Line-to-line voltages are

$$V_{ab} = V_a - V_b = 0.215 + j0.990 = 1.01\angle 77.7° \text{ p.u.}$$

$$V_{bc} = V_b - V_c = -0.215 - j0.99 + 0.215 - j0.99 = -j1.980 = 1.980\angle 270° \text{ p.u.}$$

$$V_{ca} = V_c - V_a = -0.215 + j0.99 = 1.01\angle 102.3° \text{ p.u.}$$

Since the generated voltage to neutral $E_a$ was taken as 1.0 p.u., the above line-to-line voltages are expressed in p.u. on the base of phase voltage. The postfault line voltages expressed in volts are obtained by multiplying the p.u. value with base value of phase voltage.

Line value of base voltage = 13.8 kV

$$\text{Phase value of base voltage} = \frac{13.8}{\sqrt{3}} \text{ kV} = 7.97 \text{ kV}$$

$$V_{ab} = 1.01 \times 7.97\angle 77.7° = 8.05\angle 77.7° \text{ kV}$$

$$V_{bc} = 1.98 \times 7.97\angle 270° = 15.78\angle 270° \text{ kV}$$

$$V_{ca} = 1.01 \times 7.97\angle 102.3° = 8.05\angle 102.3° \text{ kV}$$



**EXAMPLE 4.3**

Find the subtransient currents and line-to-line voltages at the fault under subtransient conditions when a line-to-line fault between phase b and c occurs at the terminals of the generator described in example 4.2. Assume that the generator is unloaded and operating at rated terminal voltage when the fault occurs. Neglect resistance.

**SOLUTION**

Let us choose the generator ratings as base values.

$$\therefore MVA_b = 20MVA \quad ; \quad kV_b = 13.8 \text{ kV}$$

$$\text{Base current, } I_b = \frac{kVA_b}{\sqrt{3}\,kV_b} = \frac{20 \times 1000}{\sqrt{3} \times 13.8}$$

$$= 836.7 \text{ A} \approx 837 \text{ A}$$

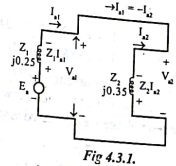



*Fig 4.3.1.*

Given that $Z_1 = j0.25$ p.u. ; $Z_2 = j0.35$ p.u. and $Z_0 = j0.1$ p.u.

For a line-to-line fault the positive and negative sequence networks are connected in parallel as shown in fig 4.3.1.

The following relations exists for a single line-to-line fault.

$$I_a = 0 \quad ; \quad I_b = -I_c \quad ; \quad I_{a0} = 0 \quad ; \quad I_{a2} = -I_{a1} \quad ; \quad V_b = V_c \quad ; \quad V_{bc} = 0$$

From fig 4.3. we get,

$$I_{a1} = \frac{E_a}{Z_1 + Z_2}$$

Here $E_a$ is phase value of induced emf. Let it be 1 p.u. (i.e. $E_a = 1\angle 0°$ p.u.)

$$\therefore I_{a1} = \frac{1}{j0.25 + j0.35} = \frac{1}{j0.6} = -j1.667 = 1.667\angle -90° \text{ p.u.}$$

$$I_{a2} = -I_{a1} = j1.667 \text{ p.u.}$$

$$I_{a0} = 0$$

The line currents are related to symmetrical components by the following equation.

$$\begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix}$$

where $a = 1\angle 120° = -0.5 + j0.866$

and $a^2 = 1\angle 240° = -0.5 - j0.866$

$$I_a = I_{a0} + I_{a1} + I_{a2} = -j1.667 + j1.667 = 0$$

$$I_b = I_{a0} + a^2 I_{a1} + a I_{a2}$$

$$= -j1.667(-0.5 - j0.866) + j1.667(-0.5 + j0.866)$$

$$= j0.833 - 1.443 - j0.833 - 1.443 = -2.886 \text{ p.u.}$$

$$I_c = I_{a0} + a I_{a1} + a^2 I_{a2} = -a I_{a2} - a^2 I_{a1} = -(a^2 I_{a1} + a I_{a2}) = -I_b = 2.886 \text{ p.u.}$$

Actual values of line currents are obtained by multiplying the p.u. values with base current.

$$I_a = 0$$

$$I_b = -2.886 \times 837 = 2416\angle 180° \text{ A}$$

$$I_c = 2.886 \times 837 = 2416\angle 0° \text{ A}$$

The fault current, $I_f = |I_b| = 2416 \text{ A} = 2.416 \text{ kA}$

From the sequence networks of the generator the symmetrical components of phase-a voltage are calculated as shown below.

$$V_{a0} = 0 \quad ; \quad V_{a1} = E_a - I_{a1} Z_1 \quad ; \quad V_{a2} = V_{a1}$$

$$\therefore V_{a1} = V_{a2} = 1 - (-j1.667)(j0.25) = 1 - 0.417 = 0.583 \text{ p.u.}$$

The phase voltages are given by,

$$\begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix}$$

$$V_a = V_{a0} + V_{a1} + V_{a2} = 0.583 + 0.583 = 1.166\angle 0° \text{ p.u.}$$

$$V_b = V_{a0} + a^2 V_{a1} + a V_{a2}$$

$$V_b = 0.583(-0.5 - j0.866) + 0.583(-0.5 + j0.866) = -0.583 \text{ p.u.}$$

$$V_c = V_b = -0.583 \text{ p.u.}$$

Line voltages are

$$V_{ab} = V_a - V_b = 1.166 + 0.583 = 1.749\angle 0° \text{ p.u.}$$

$$V_{bc} = V_b - V_c = -0.583 + 0.583 = 0 \text{ p.u.}$$

$$V_{ca} = V_c - V_a = -0.583 - 1.166 = 1.749\angle 180° \text{ p.u.}$$

Since the generated phase voltage $E_a$ is taken as 1 p.u., the above line voltages are expressed in p.u. on the base of phase voltage. The post fault line voltages expressed in volts are obtained by multiplying the p.u. value with the base value of phase voltage.

Line value of base voltage = 13.8 kV

$$\text{Phase value of base voltage} = \frac{13.8}{\sqrt{3}} = 7.9 \text{ kV}$$

$$V_{ab} = 1.749\angle 0° \times 7.97 = 13.94\angle 0° \text{ kV} \quad ; \quad V_{bc} = 0 \text{ kV}$$

$$V_{ca} = 1.749\angle 180° \times 7.97 = 13.94\angle 180° \text{ kV}$$



## EXAMPLE 4.4

Find the subtransient currents and the line-to-line voltages at the fault under subtransient conditions when a double line-to-ground fault involving phase b and c occurs at the terminals of the generator described in example 4.2. Assume that the generator is unloaded and operating at rated voltage when the fault occurs. Neglect resistance.

## SOLUTION

Let us choose the generator ratings as base values.

$$\therefore \text{MVA}_b = 20\text{MVA} \quad ; \quad \text{kV}_b = 13.8 \text{ kV}$$

$$\text{Base current, } I_b = \frac{\text{kVA}_b}{\sqrt{3}\,\text{kV}_b} = \frac{20 \times 10^3}{\sqrt{3} \times 138} = 836.7\text{A} \approx 837\text{A}$$

Given that $Z_1 = j0.25$ p.u. ; $Z_2 = j0.35$ p.u. and $Z_0 = j0.1$ p.u.

For a double line-to-ground fault the positive, negative and zero sequence networks are connected in parallel as shown in fig 4.4.1.

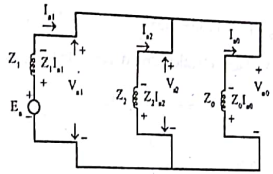



*Fig 4.4.1*

The following relations exists for a double line-to-ground fault.

$$I_{a1} = -(I_{a2} + I_{a0}) \quad ; \quad V_{a0} = V_{a1} = V_{a2} \quad ; \quad V_b = V_c = 0$$

From fig 4.4.1 we can calculate the symmetrical components of phase-a voltage and current as shown below.

$$I_{a1} = \frac{E_a}{Z_1 + Z_2 Z_0 / (Z_2 + Z_0)} = \frac{1.0 + j0}{j0.25 + (j0.35 \times j0.10)/(j0.35 + j0.10)}$$

$$= \frac{1.0}{j0.25 + j0.0778} = \frac{1.0}{j0.3278} = -j3.05 \text{ p.u.}$$

$$V_{a1} = V_{a0} = V_{a2} = E_a - I_{a1} Z_1 = 1 - (-j3.05)(j0.25) = 1.0 - 0.763 = 0.237 \text{ p.u.}$$

$$I_{a2} = \frac{-V_{a2}}{Z_2} = -\frac{0.237}{j0.35} = j0.68 \text{ p.u}$$

$$I_{a0} = \frac{-V_{a0}}{Z_0} = \frac{-0.237}{j0.10} = j2.37 \text{ p.u}$$

The line currents are related to symmetrical components by the following equation.

$$\begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix}$$

$I_a = I_{a0} + I_{a1} + I_{a2} = j2.37 - j3.05 + j0.68 = 0$

$I_b = I_{a0} + a^2 I_{a1} + a I_{a2}$

$= j2.37 + (-0.5 - j0.866)(-j3.05) + (-0.5 + j0.866)(j0.68)$

$= j1.525 - 2.641 - j0.34 - 0.589 + j2.37 = -3.230 + j3.555 = 4.80\angle 132.3°$ p.u.

$I_c = I_{a0} + a I_{a1} + a^2 I_{a2}$

$= j2.37 + (-0.5 + j0.866)(-j3.05) + (-0.5 - j0.866)(j0.68)$

$= j1.525 + 2.641 - j0.34 + 0.589 + j2.37 = 3.230 + j3.555$

$= 4.80\angle 47.7°$ p.u.

Fault current, $I_f = I_b + I_c = -3.230 + j3.555 + 3.230 + j3.555 = j7.11$ p.u.

The phase and line voltages are calculated as shown below

$V_a = V_{a1} + V_{a2} + V_{a0} = 3V_{a1} = 3 \times 0.237 = 0.711$ p.u.

$V_b = V_c = 0$

$V_{ab} = V_a - V_b = 0.711$ p.u.

$V_{bc} = 0$

$V_{ca} = V_c - V_a = -0.711$ p.u.

The actual values of line currents and voltages are calculated as shown below.

$I_a = 0$

$I_b = 837 \times 4.8\angle 132.3° = 4017\angle 132.3°$ A

$I_c = 837 \times 4.8\angle 47.7° = 4017\angle 47.7°$ A

$I_f = 837 \times 7.11\angle 90° = 5951\angle 90°$ A

$V_{ab} = 0.711 \times 7.97 = 5.66\angle 0°$ kV

$V_{bc} = 0$

$V_{ca} = -0.711 \times 7.97 = 5.66\angle 180°$ kV



**EXAMPLE 4.5**

A generator of negligible resistance having 1 p.u. voltage behind transient reactance is subjected to different types of faults.

| Type of fault | Resulting fault current in pu |
|---|---|
| 3 - phase | 3.33 |
| L - L | 2.23 |
| L - G | 3.01 |

Calculate the per unit value of 3 sequence reactances.

**SOLUTION**

**Case( i) : 3-phase fault**

The circuit model used to calculate the transient fault current for a 3-phase fault on a generator is shown in fig 4.5.1

Given that, $E_g' = 1$ p.u., $I' = 3.33$ p.u.

When load current is neglected, $E_g'' = E_g' = E_g = E_a$

$\therefore X_d' = \frac{|E_a|}{|I'|} = \frac{1}{3.33} = 0.3$ p.u.

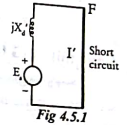


*Fig 4.5.1*

We know that the reactance during symmetrical fault is positive sequence reactance.

$\therefore$ positive sequence reactance of generator, $X_1 = X_d' = 0.3$ p.u.

**Case (ii) : L - L fault**

For a line-to-line fault on a generator the positive and negative sequence networks can be connected in parallel as shown in fig 4.5.2

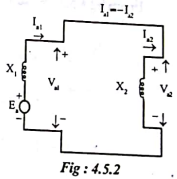


*Fig : 4.5.2*

From fig 4.5.2 we get

$$I_{a1} = \frac{E_a}{jX_1 + jX_2}$$

$$\therefore |I_{a1}| = \frac{E_a}{X_1 + X_2} \quad \text{.....(4.5.1)}$$

Let us assume that the line-to-line fault is due to a short between phase-b and phase-c. Hence the fault current is $I_b$.

From the symmetrical components of the current we can write

$I_b = I_{a0} + a^2 I_{a1} + a I_{a2}$

For a line-to-line fault, $I_{a0}=0$ and $I_{a2}=-I_{a1}$

$$\therefore I_b = 0 + a^2 I_{a1} - a I_{a1} = I_{a1}(a^2 - a)$$
$$= I_{a1}(-0.5 - j0.866 - (-0.5 + j0.866)) = I_{a1}(-j1.732) = -j1.732\, I_{a1}$$

Fault current, $I_f = |I_b| = 1.732\,|I_{a1}|$

$$\therefore |I_{a1}| = \frac{I_f}{1.732}$$

Given that $I_f = 2.23$ p.u.

$$\therefore |I_{a1}| = \frac{2.23}{1.732} = 1.2875 \text{ p.u.} \qquad \text{.....(4.5.2)}$$

From equation (4.5.1) & (4.5.2) we can write,

$$1.2875 = \frac{E_a}{X_1 + X_2}$$

$$\therefore X_1 + X_2 = \frac{E_a}{1.2875} \quad ; \quad \therefore X_2 = \frac{E_a}{1.2875} - X_1 = \frac{1}{1.2875} - 0.3 = 0.48 \text{ p.u}$$

**Case (iii) L-G fault**

For a single line-to-ground fault, on a generator the sequence nerworks are connected in series as shown in fig 4.5.3. For a single line-to-ground fault, the fault current is $|I_a|$ and it is equal to $3|I_{a1}|$.

Given that, $I_f = 3.01$ p.u.

$$\therefore 3|I_{a1}| = I_f = 3.01 \quad \text{or} \quad |I_{a1}| = \frac{3.01}{3} = 1.0033 \text{ p.u.}$$

From fig 4.5.3 we can write

$$I_{a1} = \frac{E_a}{jX_1 + jX_2 + jX_3}$$

$$\therefore |I_{a1}| = \frac{E_a}{X_1 + X_2 + X_3}$$

$$X_1 + X_2 + X_3 = \frac{E_a}{|I_{a1}|}$$

$$X_3 = \frac{E_a}{|I_{a1}|} - X_1 - X_2 = \frac{1}{1.0033} - 0.3 - 0.48 = 0.22 \text{ p.u.}$$

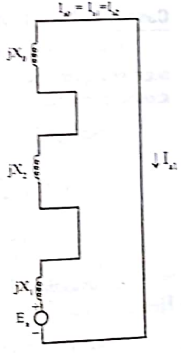



*Fig 4.5.3*

**RESULT**

Positive sequence reactance of generator = 0.3 p.u.
Negative sequence reactance of generator = 0.48 p.u.
Zero sequence reactance of generator = 0.22 p.u.



## 4.7 SHORT QUESTIONS AND ANSWERS

1. *Name the fault in which positive, negative and zero sequence component currents are equal.*

   In single line-to-ground fault the positive, negative and zero sequence component currents are equal.

2. *Name the fault in which positive and negative sequence component currents together is equal to zero sequence current in magnitude.*

   Double line-to-ground fault

3. *Name the various unsymmetrical faults in a power system.*

   The unsymmetrical faults in a power systems are
   (i) Single line-to-ground fault
   (ii) Line-to-line fault
   (iii) Double line-to-ground fault
   (iv) Open conductor fault

4. *Define negative sequence impedance?*

   The negative sequence impedance of an equipment is the impedance offered by the equipment to the flow of negative sequence current.

5. *Name the faults which do not have zero sequence currents flowing.*

   In line to line faults the zero sequence currents do not flow.

6. *Name the faults involving ground*

   The faults involving ground are
   (i) Single line to ground fault
   (ii) Double line to ground fault
   (ii) Three phase fault

7. *Define positive sequence impedance.*

   The positive sequence impedance of an equipment is the impedance offered by the equipment to the flow of positive sequence currents.

8. *In what type of fault the positive sequence component of current is equal in magnitude but opposite in phase to negative sequence components of current?*

   Line-to-line fault

9. *In which fault the negative and zero sequence currents are absent?*

   In three phase fault the negative and zero sequence currents are absent.

10. *Write the boundary condition in single line-to-ground fault?*

    The boundary conditions are

    $V_a = 0$ ; $I_b = I_c = 0$

11. What are the boundary condition in line-to-line fault?

$$I_a = 0 \; ; \; I_b + I_c = 0 \; ; \; V_b = V_c$$

12. Write down the boundary condition in double line-to-ground fault?

$$I_a = 0 \; ; \; V_b = 0 \; ; \; V_c = 0$$

13. Give the boundary condition for the 3-phase fault.

$$I_a + I_b = I_c = 0 \; ; \; V_a = V_b = V_c = 0$$

14. Draw the connection of sequence networks for a single line-to-ground fault at the terminals of an unloaded generator.

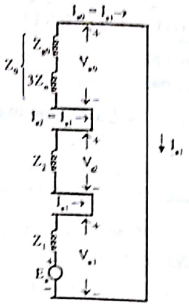


*Fig Q14 : Connection of sequence network of an unloaded generator for line-to-ground fault on phase -a*

15. Draw the connection of sequence networks for a line-to-line fault at the terminals of an unloaded generator.

16. Draw the connection of sequence networks for a double line-to-ground fault at the terminals of an unloaded generator.

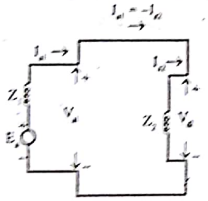


*Fig Q15 : Connection of the sequence networks of an unloaded generator for a line-to-line fault between phases b and c*

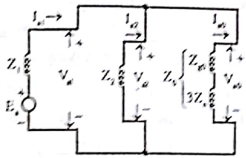


*Fig Q16 : Connection of the sequence networks of an unloaded generator for a double line-to-ground fault on phase b and c.*

17. Draw the connection of seqence networks for line-to-line fault without fault impedance.

18. Draw the connection of sequence networks for single line-to-ground fault without fault impedance?

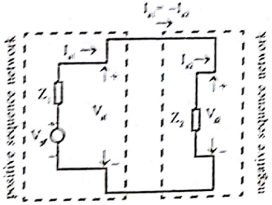



*Fig Q17 : Connection of the sequence networks for a line-to-line fault in a power system.*



19. Draw the connection of seqence networks for double line-to-ground fault without fault impedance.

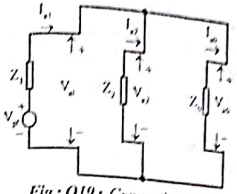


*Fig : Q19 : Connection of the sequence networks for a double line-to-ground fault in a power system.*

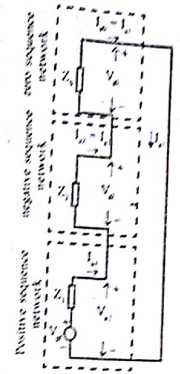



*Fig Q18 : Connection of sequence networks for a single line-to-ground fault in a power system*

20. Draw the connection of sequence networks for single line-to-ground fault through an impedance $Z_f$

21. Draw the connection of sequence networks for line-to-line fault through an impedance $Z_f$

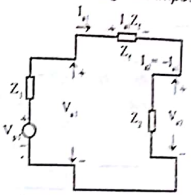


*Fig Q21 : Connection of sequence networks for a line-to-line fault through an impedance $Z_f$*

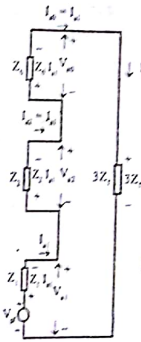


*Fig Q20 Connection of sequence networks for a single line to ground fault through an impedance $Z_f$*

22. Draw the connection of sequence networks for double line-to-ground fault through an impedance $Z_f$

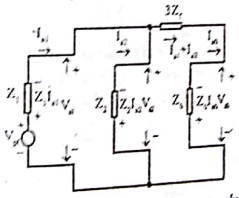


*Fig Q22 : Connection of sequence networks for a double line-to-ground fault through an impedance*